



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Test Design Specifications

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1.0 Introduction

1.1 Purpose

This Test Design Specification aims to:

- Identify the features to be tested in HMS
- Refine the testing approach based on the Test Plan
- Define the test design techniques used
- Identify test conditions and test coverage

1.2 Scope

This Test Design Specification covers the major functional features of the Hospital Management System (HMS), including:

1. User Authentication

- Register Account (F001)
- Login (F002)
- Logout (F003)

2. Patient Functions

- Book Appointment (F004)
- View Appointment History (F005)
- Cancel Appointment (F006)
- View Prescription (F007)
- Submit Contact Query (F008)

3. Doctor Functions

- View Appointment (F009)
- Prescribe Medication (F010)
- Cancel Appointment (F011)
- View Prescription List (F012)

4. Admin Functions

- Add Doctor (F013)
- Delete Doctor (F014)
- View & Search Doctor List (F015)
- View & Search Patient List (F016)
- View & Search Appointment Details (F017)
- View Prescription List (F018)
- View & Search User Queries (F019)

1.3 References

1. HMS Software Requirements Specification
2. Client challenge. (n.d.). <https://www.scribd.com/document/514605221/03-TAB-6-ATMS-Test-Design-Specification-Set-A-v1-0-0-pdf>
3. Kishan. (n.d.). GitHub - kishan0725/Hospital-Management-System: Hospital Management System using php and mysql. GitHub.
<https://github.com/kishan0725/Hospital-Management-System.git>

2.0 Test Design

2.1 Features to Be Tested

The following features are selected from the HMS Software Requirements Specification for test design. These features are categorized based on system roles and their importance.

Function ID	Function Name	Role	Risk Level
F001	Register Account	Patient	High
F002	Login	All	High
F003	Logout	All	Medium
F004	Book Appointment	Patient	High
F005	View Appointment History	Patient	Medium
F006	Cancel Appointment	Patient	Medium
F007	View Prescription	Patient	High
F008	Submit Contact Query	Patient	Low
F009	View Appointment	Doctor	High
F010	Prescribe Medication	Doctor	High
F011	Cancel Appointment	Doctor	Medium
F012	View Prescription List	Doctor	Medium
F013	Add Doctor	Admin	High
F014	Delete Doctor	Admin	High
F015	View & Search Doctor List	Admin	Medium
F016	View & Search Patient List	Admin	Medium
F017	View & Search Appointment Details	Admin	High
F018	View Prescription List	Admin	Medium
F019	View & Search User Queries	Admin	Low

Table 2.1 Features to Be Tested

2.2 Approach Refinements

As defined in the Test Plan, the following test design techniques are applied based on the nature of system features:

- i. Equivalence Partitioning
- ii. Boundary Value Analysis
- iii. Decision Table Testing
- iv. State Transition Testing
- v. Use Case Testing

These techniques are selected to ensure complete coverage of:

- Input validation
- Decision logic
- System states
- User interaction scenarios

2.2.1 F001 Register Account

2.2.1.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-01-001	All inputs are valid	TCOV-01-001	Valid input (Registration success)	First Name: Jason, Last Name: Chin, Email: jason@gmail.com , Phone: 0123456789, Password: abc123, Confirm: abc123, Gender: Male
TCON-01-002	Email format is invalid	TCOV-01-002	Invalid email (Error message)	Email: jasongmail.com
TCON-01-003	Password less than 6 characters	TCOV-01-003	Invalid password length (Error)	Password: abc12
TCON-01-004	Password and confirm password do not match	TCOV-01-004	Password mismatch (Error)	Password: abc123, Confirm: abc124
TCON-01-005	Required field is empty	TCOV-01-005	Missing input (Validation error)	First Name: (empty)

TCON-01-006	Phone number format invalid	TCOV-01-006	Invalid phone input (Error)	Phone: abc123
TCON-01-007	Gender not selected	TCOV-01-007	Missing selection (Validation error)	Gender: (not selected)

Table 2.2.1.1 Register Account Equivalence Partitioning Test Conditions & Coverage

2.2.1.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-01-008	Password = 0 characters	TCOV-01-008	Password = 0 characters (Invalid)	No input
TCON-01-009	Password = 5 characters	TCOV-01-009	Password < 6 characters (Invalid)	abc12
TCON-01-010	Password = 6 characters	TCOV-01-010	Password = 6 characters (Valid)	abc123
TCON-01-011	Password = 7 characters	TCOV-01-011	Password > 6 characters (Valid)	abc1234
TCON-01-012	Valid password	TCOV-01-012	Valid password length (Valid output)	abc123
TCON-01-013	Invalid password	TCOV-01-013	Invalid password length (Invalid output)	abc12

Table 2.2.1.2 Register Account Boundary Value Analysis Test Conditions & Coverage

2.2.1.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-01-014	All inputs valid
TCON-01-015	Email valid
TCON-01-016	Password valid
TCON-01-017	Password match
TCON-01-018	Missing field
TCON-01-019	Invalid email
TCON-01-020	Invalid password
TCON-01-021	Password mismatch

Table 2.2.1.3a Register Account Decision Table Test Conditions

Condition	TCOV-01-014	TCOV-01-015	TCOV-01-016	TCOV-01-017
Email Valid	Y	N	Y	Y
Password Valid (≥6)	Y	Y	N	Y
Password Match	Y	Y	Y	N
All Fields Filled	Y	Y	Y	Y
Output				
Register Success	Y	N	N	N
Show Error Message	N	Y	Y	Y

Table 2.2.1.3b Register Account Decision Table

2.2.1.4 State Transition Testing

State	Enter Details (ED)	Submit Form (SF)	Valid Input (VI)	Invalid Input (II)	Password Mismatch (PM)
Registration Page (S1)	S2/-	S1/-	S1/-	S1/-	S1/-
Input Details (S2)	S2/-	S3/Validate	S2/-	S2/-	S2/-
Validation (S3)	S3/-	S3/-	S4/Register Success	S5/Error Message	S5/Not Matching
Registered (S4)	S4/-	S4/-	S4/-	S4/-	S4/-
Error Displayed (S5)	S2/Re-enter Details	S5/-	S5/-	S5/-	S5/-

Table 2.2.1.4a Register Account State Table

Test Coverage ID	Test Coverage
TCOV-01-018	S1 to S2 with input ED
TCOV-01-019	S2 to S3 with input SF
TCOV-01-020	S3 to S4 with input VI
TCOV-01-021	S3 to S5 with input II
TCOV-01-022	S3 to S5 with input PM
TCOV-01-023	S5 to S2 with input ED

Table 2.2.1.4b Register Account State Coverage**2.2.1.5 Use Case Testing**

Use Case ID	UC001
Use Case	F001 Register Account
Purpose	To allow a patient to create a new account.
Actor	Patient
Trigger	Patient opens the home page and enters registration details.
Precondition	Patient is not logged in and the registration form is available.
Main Flow	1. Patient accesses the home page. 2. System displays the registration form. 3. Patient enters first name, last name, email, phone number, password, confirm password, and gender. 4. Patient clicks "Register" button. 5. System validates all input fields. 6. System creates a new patient account. 7. System redirects patient to the dashboard.
Alternate Flow – Empty Field	1.1 Patient leaves one or more required fields empty. 1.2 System displays validation message "Please fill all required fields". 1.3 Back to Main Flow Step 3.
Alternate Flow – Invalid Email	2.1 Patient enters an invalid email format. 2.2 System displays error message "Invalid email format". 2.3 Back to Main Flow Step 3.
Alternate Flow – Password Mismatch	3.1 Patient enters password and confirm password that do not match. 3.2 System displays error message "Password not matching". 3.3 Back to Main Flow Step 3.
Alternate Flow – Invalid	4.1 Patient enters password less than 6 characters.

Password Length	4.2 System displays error message “Password must be at least 6 characters”. 4.3 Back to Main Flow Step 3.
Rules	i. All required fields must be filled. ii. Password must be at least 6 characters. iii. Password and confirm password must match.
Note	Registration is only available for patient users.

Table 2.2.1.5a Register Account Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-01-024	Main flow	TCOV-01-024	Successful patient registration	First Name: John, Last Name: Lee, Email: john@gmail.com, Phone: 0123456789, Password: abc123, Confirm Password: abc123, Gender: Male
TCON-01-025	Alternate Flow – Required field empty	TCOV-01-025	Missing input validation	First Name: empty
TCON-01-026	Alternate Flow – Password mismatch	TCOV-01-026	Password and confirm password not matching	Password: abc123, Confirm Password: abc124
TCON-01-027	Alternate Flow – Password fewer than 6 characters	TCOV-01-027	Invalid password length	Password: abc12
TCON-01-028	Alternate Flow – Invalid email format	TCOV-01-028	Invalid email validation	Email: johngmail.com

Table 2.2.1.5b Register Account Use Case Test Conditions & Coverage

2.2.2 F002 Login

2.2.2.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-02-001	Valid registered patient email and correct password	TCOV-02-001	Valid patient login	Email: patient@gmail.com, Password: abc123
TCON-02-002	Valid doctor username and correct password	TCOV-02-002	Valid doctor login	Username: doctor01, Password: doc123
TCON-02-003	Valid admin username and correct password	TCOV-02-003	Valid admin login	Username: admin, Password: admin123
TCON-02-004	Account does not exist	TCOV-02-004	Invalid account login rejected	Email/Username: unknown@gmail.com
TCON-02-005	Password is incorrect	TCOV-02-005	Invalid password rejected	Password: wrong123
TCON-02-006	Required field is empty	TCOV-02-006	Missing input validation	Email/Username: empty, Password: empty

Table 2.2.2.1 Login Equivalence Partitioning Test Conditions & Coverage

2.2.2.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-02-007	Password = 0 characters	TCOV-02-007	Password empty input (Invalid)	No input
TCON-02-008	Password = 5 characters	TCOV-02-008	Password less than 6 characters (Invalid)	abc12
TCON-02-009	Password = 6 characters	TCOV-02-009	Password minimum length accepted if correct (Valid)	abc123

TCON-02-010	Password = 7 characters	TCOV-02-010	Password more than minimum length accepted if correct (Valid)	abc1234
TCON-02-011	Valid password	TCOV-02-011	Correct password allows login	abc123
TCON-02-012	Invalid password	TCOV-02-012	Incorrect password rejects login	wrong12

Table 2.2.2.2 Login Boundary Value Analysis Test Conditions & Coverage

2.2.2.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-02-013	Account exists
TCON-02-014	Account does not exist
TCON-02-015	Password correct
TCON-02-016	Password incorrect
TCON-02-017	Role is Patient
TCON-02-018	Role is Doctor
TCON-02-019	Role is Admin
TCON-02-020	Empty input

Table 2.2.2.3a Login Decision Table Test Conditions

Condition	TCOV-02-013	TCOV-02-014	TCOV-02-015	TCOV-02-016	TCOV-02-017
Account Exists	Y	Y	Y	N	-
Password Correct	Y	Y	Y	-	-
Role = Patient	Y	N	N	-	-
Role = Doctor	N	Y	N	-	-
Role = Admin	N	N	Y	-	-
Input Empty	N	N	N	N	Y
Output					
Redirect to Patient Dashboard	Y	N	N	N	N
Redirect to Doctor Dashboard	N	Y	N	N	N

Redirect to Admin Dashboard	N	N	Y	N	N
Show Error Message	N	N	N	Y	Y

Table 2.2.2.3b Login Decision Table

2.2.2.4 State Transition Testing

State	Enter Credentials (EC)	Submit Login (SL)	Valid Patient (VP)	Valid Doctor (VD)	Valid Admin (VA)	Invalid Login (IL)	Empty Input (EI)
Login Page (S1)	S2/-	S1/-	S1/-	S1/-	S1/-	S1/-	S1/-
Credentials Entered (S2)	S2/-	S3/Validate	S2/-	S2/-	S2/-	S2/-	S2/-
Validation (S3)	S3/-	S3/-	S4/Patient Dashboard	S5/Doctor Dashboard	S6/Admin Dashboard	S7/Error Message	S7/Validation Message
Patient Dashboard (S4)	S4/-	S4/-	S4/-	S4/-	S4/-	S4/-	S4/-
Doctor Dashboard (S5)	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-
Admin Dashboard (S6)	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-
Error Displayed (S7)	S2/Re-enter Credentials	S7/-	S7/-	S7/-	S7/-	S7/-	S7/-

Table 2.2.2.4a Login State Table

Test Coverage ID	Test Coverage
TCOV-02-018	S1 to S2 with input EC
TCOV-02-019	S2 to S3 with input SL
TCOV-02-020	S3 to S4 with input VP
TCOV-02-021	S3 to S5 with input VD
TCOV-02-022	S3 to S6 with input VA
TCOV-02-023	S3 to S7 with input IL
TCOV-02-024	S3 to S7 with input EI
TCOV-02-025	S7 to S2 with input EC

Table 2.2.2.4b Login State Coverage**2.2.2.5 Use Case Testing**

Use Case ID	UC002
Use Case	F002 Login
Purpose	To allow patients, doctors, and administrators to log into the system using valid credentials.
Actor	Patient, Doctor, Admin
Trigger	User accesses the login page and enters login credentials.
Precondition	User is not logged into the system and a valid account already exists.
Main Flow	1. User accesses the home page. 2. System displays login options for Patient, Doctor, and Admin. 3. User selects the correct login interface. 4. User enters email/username and password. 5. User clicks the Login button. 6. System validates the entered credentials. 7. System grants access. 8. System redirects user to the corresponding dashboard.
Alternate Flow – Empty Field	1.1 User leaves username/email or password field empty. 1.2 System displays a validation message. 1.3 Back to Main Flow Step 4.
Alternate Flow – Invalid Credentials	2.1 User enters incorrect username/email or password. 2.2 System denies login. 2.3 System displays an error message. 2.4 Back to Main Flow Step 4.
Alternate Flow – Wrong Role Login	3.1 User enters credentials in the wrong role login interface. 3.2 System rejects login. 3.3 System displays an error message. 3.4 Back to Main Flow Step 3.
Rules	i. A valid account must exist before login. ii. User must enter correct credentials for the selected role. iii. System shall redirect user to the correct dashboard after successful login.
Note	Patient uses email and password, while Doctor and Admin use username and password.

Table 2.2.2.5a Login Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
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TCON-02-021	Main flow – Patient login	TCOV-02-026	Successful patient login	Email: patient@gmail.com, Password: abc123
TCON-02-022	Main flow – Doctor login	TCOV-02-027	Successful doctor login	Username: doctor01, Password: doc123
TCON-02-023	Main flow – Admin login	TCOV-02-028	Successful admin login	Username: admin, Password: admin123
TCON-02-024	Alternate Flow – Empty field	TCOV-02-029	Missing input validation	Username/Email: empty, Password: empty
TCON-02-025	Alternate Flow – Invalid credentials	TCOV-02-030	Login rejected	Username/Email: patient@gmail.com, Password: wrong123
TCON-02-026	Alternate Flow – Wrong role login	TCOV-02-031	Login rejected due to wrong role	Doctor username entered in Patient login page

Table 2.2.2.5b Login Use Case Test Conditions & Coverage

2.2.3 F003 Logout

2.2.3.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-03-001	User has active session	TCOV-03-001	Valid logout (session terminated)	Logged-in patient clicks Logout
TCON-03-002	User has no active session	TCOV-03-002	Invalid logout access handled	Directly access logout page without login
TCON-03-003	User clicks “Back to Login Page” after logout	TCOV-03-003	Redirect to login page	Click Back to Login Page button

Table 2.2.3.1 Logout Equivalence Partitioning Test Conditions & Coverage

2.2.3.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-03-004	Logout clicked immediately after login	TCOV-03-004	Minimum active session duration logout	Login then click Logout immediately

TCON-03-005	Logout clicked after normal session usage	TCOV-03-005	Normal active session logout	Login, open dashboard, click Logout
TCON-03-006	Logout attempted after session already ended	TCOV-03-006	Expired/terminated session handled	Click browser back after logout

Table 2.2.3.2 Logout Boundary Value Analysis Test Conditions & Coverage

2.2.3.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-03-007	User logged in
TCON-03-008	Logout button clicked
TCON-03-009	Session active
TCON-03-010	Session not active

Table 2.2.3.3a Logout Decision Table Test Conditions

Condition	TCOV-03-007	TCOV-03-008	TCOV-03-009
User Logged In	Y	Y	N
Logout Button Clicked	Y	N	-
Session Active	Y	Y	N
Output			
Terminate Session	Y	N	N
Display "You have logged out"	Y	N	N
Redirect/Remain on Current Page	N	Y	N
Show Login Page / Access Denied	N	N	Y

Table 2.2.3.3b Logout Decision Table

2.2.3.4 State Transition Testing

State	Click Logout (CL)	Session Terminated (ST)	Click Back to Login (BL)	No Active Session (NS)
Dashboard (S1)	S2/Terminate Session	S1/-	S1/-	S4/Login Page
Logout Page (S2)	S2/-	S3/Display Message	S4/Login Page	S4/Login Page

Logged Out Message (S3)	S3/-	S3/-	S4/Login Page	S4/Login Page
Login Page (S4)	S4/-	S4/-	S4/-	S4/-

Table 2.2.3.4a Logout State Table

Test Coverage ID	Test Coverage
TCOV-03-010	S1 to S2 with input CL
TCOV-03-011	S2 to S3 with input ST
TCOV-03-012	S3 to S4 with input BL
TCOV-03-013	S1 to S4 with input NS

Table 2.2.3.4b Logout State Coverage

2.2.3.5 Use Case Testing

Use Case ID	UC003
Use Case	F003 Logout
Purpose	To allow users to exit the system and terminate the current session.
Actor	Patient, Doctor, Admin
Trigger	User clicks Logout button from the corresponding dashboard.
Precondition	User is logged into the system and is currently on the patient, doctor, or admin dashboard.
Main Flow	1. User clicks the Logout button. 2. System terminates the current user session. 3. System redirects user to the logout page. 4. System displays “You have logged out.” 5. System displays “Back to Login Page” button. 6. User clicks the button to return to the login page.
Alternate Flow – No Active Session	1.1 User tries to access logout page without an active session. 1.2 System redirects user to the login page or prevents dashboard access.
Rules	i. System must terminate the user session after logout. ii. Logged-out users must not be able to access dashboard pages using the browser back button.
Note	Logout function applies to Patient, Doctor, and Admin users.

Table 2.2.3.5a Logout Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
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TCON-03-011	Main flow – Patient logout	TCOV-03-014	Patient session terminated	Logged-in patient clicks Logout
TCON-03-012	Main flow – Doctor logout	TCOV-03-015	Doctor session terminated	Logged-in doctor clicks Logout
TCON-03-013	Main flow – Admin logout	TCOV-03-016	Admin session terminated	Logged-in admin clicks Logout
TCON-03-014	Alternate Flow – No active session	TCOV-03-017	Logout access handled without active session	Direct logout URL access without login
TCON-03-015	Browser back after logout	TCOV-03-018	Dashboard access blocked after logout	Click browser back after logout

Table 2.2.3.5b Logout Use Case Test Conditions & Coverage

2.2.4 F004 Book Appointment

2.2.4.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-04-001	All appointment inputs are valid	TCOV-04-001	Appointment booking success	Specialization: Cardiologist, Doctor: Dr. John, Date: future date, Time: 10:00
TCON-04-002	Specialization is not selected	TCOV-04-002	Missing specialization validation	Specialization: not selected
TCON-04-003	Doctor is not selected	TCOV-04-003	Missing doctor validation	Doctor: not selected
TCON-04-004	Appointment date is not selected	TCOV-04-004	Missing date validation	Date: empty
TCON-04-005	Appointment time is not selected	TCOV-04-005	Missing time validation	Time: not selected
TCON-04-006	Appointment date/time is not in the future	TCOV-04-006	Invalid date/time rejected	Date: yesterday, Time: 08:00
TCON-04-007	Doctor is unavailable for selected date/time	TCOV-04-007	Unavailable doctor rejected	Doctor: Dr. John, Date: booked date, Time: booked time

Table 2.2.4.1 Book Appointment Equivalence Partitioning Test Conditions & Coverage

2.2.4.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-04-008	Appointment date = past date	TCOV-04-008	Past date is invalid	Date: yesterday
TCON-04-009	Appointment date = current date/time already passed	TCOV-04-009	Current date with past time is invalid	Date: today, Time: earlier than current time
TCON-04-010	Appointment date = future date	TCOV-04-010	Future date is valid	Date: tomorrow
TCON-04-011	Appointment time = first available slot	TCOV-04-011	Lower boundary time slot is valid	Time: 08:00
TCON-04-012	Appointment time = last available slot	TCOV-04-012	Upper boundary time slot is valid	Time: 16:00
TCON-04-013	Appointment time outside allowed slot	TCOV-04-013	Invalid time slot rejected	Time: 18:00

Table 2.2.4.2 Book Appointment Boundary Value Analysis Test Conditions & Coverage

2.2.4.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-04-014	Patient is logged in
TCON-04-015	All required fields selected
TCON-04-016	Appointment date/time is in the future
TCON-04-017	Doctor is available
TCON-04-018	Doctor is unavailable
TCON-04-019	Required field is missing

Table 2.2.4.3a Book Appointment Decision Table Test Conditions

Condition	TCOV-04-014	TCOV-04-015	TCOV-04-016	TCOV-04-017
Patient Logged In	Y	Y	Y	Y
All Required Fields Selected	Y	N	Y	Y
Date/Time in Future	Y	-	N	Y
Doctor Available	Y	-	-	N
Output				

Create Appointment Record	Y	N	N	N
Display “Your appointment successfully booked”	Y	N	N	N
Display Validation Message	N	Y	N	N
Display “Select a time or date in the future!”	N	N	Y	N
Display Doctor Unavailable Message	N	N	N	Y

Table 2.2.4.3b Book Appointment Decision Table

2.2.4.4 State Transition Testing

State	Select Booking Function (SBF)	Select Specialization (SS)	Select Doctor (SD)	Select Date/Time (SDT)	Submit Booking (SB)
Patient Dashboard (S1)	S2/Display Booking Page	S1/-	S1/-	S1/-	S1/-
Booking Page (S2)	S2/-	S3/Load Doctors	S2/-	S2/-	S2/-
Doctor Selected (S3)	S3/-	S3/-	S4/Display Fee	S3/-	S3/-
Date/Time Selected (S4)	S4/-	S4/-	S4/-	S5/Ready to Submit	S4/-
Validation (S5)	S5/-	S5/-	S5/-	S5/-	S6/Validate
Appointment Created (S6)	S6/-	S6/-	S6/-	S6/-	S6/-
Success Displayed (S7)	S7/-	S7/-	S7/-	S7/-	S7/-
Error Displayed (S8)	S2/Re-enter Booking Details	S8/-	S8/-	S8/-	S8/-

Table 2.2.4.4a Book Appointment State Table

Test Coverage ID	Test Coverage
TCOV-04-018	S1 to S2 with input SBF
TCOV-04-019	S2 to S3 with input SS
TCOV-04-020	S3 to S4 with input SD
TCOV-04-021	S4 to S5 with input SDT

TCOV-04-022	S5 to S6 with input SB
TCOV-04-023	S6 to S7 with input VB
TCOV-04-024	S6 to S8 with input IB
TCOV-04-025	S8 to S2 with input re-enter booking details

Table 2.2.4.4b Book Appointment State Coverage

2.2.4.5 Use Case Testing

Field	Description
Use Case ID	UC004
Use Case	F004 Book Appointment
Purpose	To allow patients to book appointments.
Actor	Patient
Trigger	Patient selects the Book Appointment function from the dashboard.
Precondition	Patient is logged into the system.
Main Flow	1. Patient selects Book Appointment. 2. System displays appointment booking page. 3. Patient selects specialization. 4. Patient selects doctor. 5. System displays consultation fee based on selected doctor. 6. Patient selects appointment date. 7. Patient selects appointment time. 8. Patient clicks the Create New Entry button. 9. System validates the appointment information. 10. System creates a new appointment record. 11. System displays "Your appointment successfully booked".
Alternate Flow – Doctor Unavailable	1.1 Patient selects a doctor who is not available on the chosen date or time. 1.2 System displays "We are sorry to inform that the doctor is not available in this time or date. Please choose different time or date!" 1.3 Back to Main Flow Step 6.
Alternate Flow – Invalid Date or Time	2.1 Patient selects appointment date or time that is not in the future. 2.2 System displays "Select a time or date in the future!" 2.3 Back to Main Flow Step 6.
Alternate Flow – Missing Required Field	3.1 Patient does not select specialization, doctor, date, or time. 3.2 System displays a validation message. 3.3 Back to Main Flow Step 3.
Rules	i. Patient must be logged into the system. ii. Patient must select specialization, doctor, appointment date, and appointment time.

	iii. Consultation fee shall be displayed based on the selected doctor. iv. Appointment date and time must be in the future. v. Selected doctor must be available at the chosen date and time.
Note	Appointment booking is only available for logged-in patient users.

Table 2.2.4.5a Book Appointment Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-04-020	Main flow	TCOV-04-026	Successful appointment booking	Specialization: Cardiologist, Doctor: Dr. John, Date: tomorrow, Time: 10:00
TCON-04-021	Alternate Flow – Doctor unavailable	TCOV-04-027	Doctor unavailable handling	Doctor: Dr. John, Date: booked date, Time: booked time
TCON-04-022	Alternate Flow – Invalid date or time	TCOV-04-028	Past/current invalid date or time rejected	Date: yesterday, Time: 08:00
TCON-04-023	Alternate Flow – Missing required field	TCOV-04-029	Missing input validation	Specialization: not selected, Doctor: empty

Table 2.2.4.5b Book Appointment Use Case Test Conditions & Coverage

2.2.5 F005 View Appointment History

2.2.5.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-05-001	Patient is logged in and has appointment records	TCOV-05-001	Appointment history displayed	Patient account with 2 appointments
TCON-05-002	Patient is logged in but has no appointment records	TCOV-05-002	Empty history handled	Patient account with no appointments

TCON-05-003	Patient is not logged in	TCOV-05-003	Access rejected / redirected to login	Direct access appointment history URL
TCON-05-004	Appointment status is Active	TCOV-05-004	Active status displayed with cancel action	Appointment Status: Active
TCON-05-005	Appointment status is Cancelled	TCOV-05-005	Cancelled status displayed without valid cancel action	Appointment Status: Cancelled

Table 2.2.5.1 View Appointment History Equivalence Partitioning Test Conditions & Coverage

2.2.5.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-05-006	Number of appointment records = 0	TCOV-05-006	Empty appointment list displayed	0 records
TCON-05-007	Number of appointment records = 1	TCOV-05-007	Single appointment record displayed	1 record
TCON-05-008	Number of appointment records > 1	TCOV-05-008	Multiple appointment records displayed	5 records
TCON-05-009	Appointment date = past date	TCOV-05-009	Past appointment displayed in history	Date: 2026-04-01
TCON-05-010	Appointment date = future date	TCOV-05-010	Upcoming appointment displayed in history	Date: 2026-05-01

Table 2.2.5.2 View Appointment History Boundary Value Analysis Test Conditions & Coverage

2.2.5.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-05-011	Patient is logged in
TCON-05-012	Appointment records exist
TCON-05-013	Appointment records do not exist
TCON-05-014	Appointment status is Active
TCON-05-015	Appointment status is Cancelled

Table 2.2.5.3a View Appointment History Decision Table Test Conditions

Condition	TCOV-05-011	TCOV-05-012	TCOV-05-013	TCOV-05-014
Patient Logged In	Y	Y	Y	N
Appointment Records Exist	Y	Y	N	-
Appointment Status Active	Y	N	-	-
Appointment Status Cancelled	N	Y	-	-
Output				
Display Appointment History	Y	Y	N	N
Display Cancel Action	Y	N	N	N
Display Empty List / Message	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.5.3b View Appointment History Decision Table

2.2.5.4 State Transition Testing

State	Select Appointment History (SAH)	Records Found (RF)	No Records Found (NRF)	Active Appointment (AA)	Cancelled Appointment (CA)	Not Logged In (NLI)
Patient Dashboard (S1)	S2/Retrieve Records	S1/-	S1/-	S1/-	S1/-	S5/Login Page
Retrieve Appointment Records (S2)	S2/-	S3/Display History	S4/Display Empty List	S2/-	S2/-	S5/Login Page
Appointment History Displayed (S3)	S3/-	S3/-	S3/-	S3/Show Cancel Action	S3/Show Cancelled Status	S3/-
Empty History Displayed (S4)	S4/-	S4/-	S4/-	S4/-	S4/-	S4/-
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.5.4a View Appointment History State Table

Test Coverage ID	Test Coverage
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TCOV-05-015	S1 to S2 with input SAH
TCOV-05-016	S2 to S3 with input RF
TCOV-05-017	S2 to S4 with input NRF
TCOV-05-018	S3 to S3 with input AA
TCOV-05-019	S3 to S3 with input CA
TCOV-05-020	S1 to S5 with input NLI

Table 2.2.5.4b View Appointment History State Coverage

2.2.5.5 Use Case Testing

Field	Description
Use Case ID	UC005
Use Case	F005 View Appointment History
Purpose	To allow patients to view past and current appointment records.
Actor	Patient
Trigger	Patient selects the Appointment History function from the dashboard.
Precondition	Patient is logged into the system and has access to the patient dashboard.
Main Flow	1. Patient selects the Appointment History function. 2. System retrieves the patient's appointment records. 3. System displays the appointment history list. 4. Patient views appointment details including doctor name, consultation fee, appointment date, appointment time, and current status.
Alternate Flow – No Appointment Records	1.1 Patient has no appointment records. 1.2 System displays an empty list or appropriate message. 1.3 Back to Main Flow Step 1.
Alternate Flow – Active Appointment	2.1 Appointment status is Active. 2.2 System displays the Cancel button for the appointment.
Alternate Flow – Cancelled Appointment	3.1 Appointment status is Cancelled. 3.2 System displays the cancelled status and does not allow further cancellation.
Alternate Flow – Not Logged In	4.1 User attempts to access appointment history without login. 4.2 System redirects user to the login page.
Rules	i. Patient must be logged into the system. ii. Only appointments belonging to the logged-in patient shall be displayed. iii. System shall display the current status of each appointment. iv. Only Active appointments can be cancelled.

Note	Appointment history is read-only except for the Cancel action on Active appointments.
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Table 2.2.5.5a View Appointment History Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-05-016	Main flow	TCOV-05-021	Appointment history displayed	Patient with existing appointments
TCON-05-017	Alternate Flow – No appointment records	TCOV-05-022	Empty history displayed	Patient with 0 appointments
TCON-05-018	Alternate Flow – Active appointment	TCOV-05-023	Active appointment with Cancel button displayed	Appointment Status: Active
TCON-05-019	Alternate Flow – Cancelled appointment	TCOV-05-024	Cancelled appointment displayed without valid cancel action	Appointment Status: Cancelled
TCON-05-020	Alternate Flow – Not logged in	TCOV-05-025	Redirect to login page	Direct URL access without login

Table 2.2.5.5b View Appointment History Use Case Test Conditions & Coverage

2.2.6 F006 Cancel Appointment

2.2.6.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-06-001	Patient is logged in and selected appointment is active	TCOV-06-001	Appointment cancellation success	Logged-in patient, Appointment Status: Active
TCON-06-002	Patient is logged in but appointment is already cancelled	TCOV-06-002	Cancellation rejected	Appointment Status: Cancelled
TCON-06-003	Patient confirms cancellation	TCOV-06-003	Status updated to cancelled	Click OK in confirmation dialog
TCON-06-004	Patient rejects cancellation	TCOV-06-004	Appointment remains active	Click Cancel in confirmation dialog

TCON-06-005	Patient is not logged in	TCOV-06-005	Access rejected / redirected to login	Direct cancel URL access without login
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Table 2.2.6.1 Cancel Appointment Equivalence Partitioning Test Conditions & Coverage

2.2.6.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-06-006	Number of active appointments = 0	TCOV-06-006	No cancel action available	0 active appointments
TCON-06-007	Number of active appointments = 1	TCOV-06-007	One cancel action available	1 active appointment
TCON-06-008	Number of active appointments > 1	TCOV-06-008	Multiple cancel actions available	3 active appointments
TCON-06-009	Appointment status changes from Active to Cancelled	TCOV-06-009	Status update boundary covered	Active → Cancelled

Table 2.2.6.2 Cancel Appointment Boundary Value Analysis Test Conditions & Coverage

2.2.6.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-06-010	Patient is logged in
TCON-06-011	Appointment exists
TCON-06-012	Appointment is active
TCON-06-013	Appointment is already cancelled
TCON-06-014	Patient confirms cancellation
TCON-06-015	Patient rejects confirmation

Table 2.2.6.3a Cancel Appointment Decision Table Test Conditions

Condition	TCOV-06-010	TCOV-06-011	TCOV-06-012	TCOV-06-013
Patient Logged In	Y	Y	Y	N

Appointment Exists	Y	Y	Y	-
Appointment Active	Y	Y	N	-
Confirmation Accepted	Y	N	-	-
Output				
Update Status to Cancelled	Y	N	N	N
Display “Your appointment successfully cancelled”	Y	N	N	N
Keep Appointment Active	N	Y	N	N
Display Cancellation Not Allowed / Error Message	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.6.3b Cancel Appointment Decision Table

2.2.6.4 State Transition Testing

State	Select Appointment History (SAH)	Click Cancel (CC)	Confirm Cancel (CF)	Reject Cancel (RC)	Already Cancelled (AC)	Not Logged In (NLI)
Patient Dashboard (S1)	S2/Display History	S1/-	S1/-	S1/-	S1/-	S6/Login Page
Appointment History (S2)	S2/-	S3/Display Confirmation	S2/-	S2/-	S5/Error Message	S6/Login Page
Confirmation Prompt (S3)	S3/-	S3/-	S4/Update Status	S2/Return to History	S3/-	S6/Login Page
Appointment Cancelled (S4)	S4/-	S4/-	S4/-	S4/-	S4/-	S4/-
Error Displayed (S5)	S2/Return to History	S5/-	S5/-	S5/-	S5/-	S6/Login Page
Login Page (S6)	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-

Table 2.2.6.4a Cancel Appointment State Table

Test Coverage ID	Test Coverage
TCOV-06-014	S1 to S2 with input SAH
TCOV-06-015	S2 to S3 with input CC
TCOV-06-016	S3 to S4 with input CF

TCOV-06-017	S3 to S2 with input RC
TCOV-06-018	S2 to S5 with input AC
TCOV-06-019	S1 to S6 with input NLI

Table 2.2.6.4b Cancel Appointment State Coverage

2.2.6.5 Use Case Testing

Use Case ID	UC006
Use Case	F006 Cancel Appointment
Purpose	To allow a patient to cancel an active appointment.
Actor	Patient
Trigger	Patient clicks the Cancel button for an active appointment.
Precondition	Patient is logged into the system and has at least one active appointment.
Main Flow	1. Patient selects the Appointment History function. 2. System displays the patient's appointment history. 3. Patient clicks the Cancel button for an active appointment. 4. System displays confirmation message "Are you sure you want to cancel this appointment?" 5. Patient clicks OK to confirm cancellation. 6. System updates the appointment status to cancelled. 7. System displays "Your appointment successfully cancelled". 8. System redirects patient back to the dashboard page.
Alternate Flow – Confirmation Rejected	1.1 Patient clicks Cancel in the confirmation dialog. 1.2 System does not cancel the appointment. 1.3 System returns to the appointment history page.
Alternate Flow – Already Cancelled Appointment	2.1 Patient selects an appointment that is already cancelled. 2.2 System does not allow the cancellation process. 2.3 System displays an appropriate message.
Alternate Flow – Not Logged In	3.1 User attempts to cancel an appointment without logging in. 3.2 System redirects user to the login page.
Rules	i. Patient must be logged into the system. ii. Only active appointments can be cancelled. iii. System shall request confirmation before cancelling the appointment.
Note	Appointment records are updated by status change and should not be physically deleted.

Table 2.2.6.5a Cancel Appointment Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-06-016	Main flow	TCOV-06-020	Successful appointment cancellation	Logged-in patient, Appointment Status: Active, Confirm: OK
TCON-06-017	Alternate Flow – Confirmation rejected	TCOV-06-021	Cancellation aborted	Confirmation: Cancel
TCON-06-018	Alternate Flow – Already cancelled appointment	TCOV-06-022	Cancellation not allowed	Appointment Status: Cancelled
TCON-06-019	Alternate Flow – Not logged in	TCOV-06-023	Redirect to login page	Direct cancel URL access without login

Table 2.2.6.5b Cancel Appointment Use Case Test Conditions & Coverage

2.2.7 F007 View Prescription

2.2.7.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-07-001	Patient is logged in and prescription exists	TCOV-07-001	Prescription displayed successfully	Patient with prescription record
TCON-07-002	Patient is logged in but no prescription exists	TCOV-07-002	Empty prescription list displayed	Patient with no prescription
TCON-07-003	Patient is not logged in	TCOV-07-003	Access denied / redirect to login	Direct access URL
TCON-07-004	Patient clicks Pay Bill button	TCOV-07-004	Bill payment process started	Pay Bill clicked
TCON-07-005	Bill payment is successful	TCOV-07-005	"Bill Paid Successfully" displayed	Valid payment
TCON-07-006	Bill payment is not completed	TCOV-07-006	PDF bill is not generated	Payment failed
TCON-07-007	Patient generates PDF bill after payment	TCOV-07-007	Printable PDF bill generated	Paid prescription record

Table 2.2.7.1 View Prescription Equivalence Partitioning Test Conditions & Coverage

2.2.7.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-07-008	Number of prescription records = 0	TCOV-07-008	Empty list displayed	0 records
TCON-07-009	Number of prescription records = 1	TCOV-07-009	Single prescription displayed	1 record
TCON-07-010	Number of prescription records > 1	TCOV-07-010	Multiple prescriptions displayed	5 records
TCON-07-011	PDF generation before payment	TCOV-07-011	PDF generation blocked	Unpaid bill
TCON-07-012	PDF generation after payment	TCOV-07-012	PDF generation allowed	Paid bill

Table 2.2.7.2 View Prescription Boundary Value Analysis Test Conditions & Coverage

2.2.7.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-07-013	Patient is logged in
TCON-07-014	Prescription exists
TCON-07-015	Bill payment successful
TCON-07-016	Bill payment failed
TCON-07-017	Generate PDF requested

Table 2.2.7.3a View Prescription Decision Table Test Conditions

Condition	TCOV-07-013	TCOV-07-014	TCOV-07-015	TCOV-07-016
Patient Logged In	Y	Y	Y	N
Prescription Exists	Y	Y	N	-
Bill Payment Successful	Y	N	-	-
Generate PDF Requested	Y	Y	Y	-
Output				
Display Prescription	Y	Y	N	N
Display "Bill Paid Successfully"	Y	N	N	N
Generate PDF Bill	Y	N	N	N

Do Not Generate PDF	N	Y	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.7.3b View Prescription Decision Table

2.2.7.4 State Transition Testing

State	Select Prescription (SP)	Pay Bill (PB)	Payment Success (PS)	Payment Failed (PF)	Generate PDF (GP)
Patient Dashboard (S1)	S2/Retrieve Prescription	S1/-	S1/-	S1/-	S1/-
Prescription Displayed (S2)	S2/-	S3/Process Payment	S2/-	S2/-	S2/-
Payment Processing (S3)	S3/-	S3/-	S4/Payment Success	S5/Payment Failed	S3/-
Bill Paid (S4)	S4/-	S4/-	S4/-	S4/-	S7/Generate PDF
Payment Failed (S5)	S2/Back	S5/-	S5/-	S5/-	S5/-
Login Page (S6)	S6/-	S6/-	S6/-	S6/-	S6/-
PDF Generated (S7)	S7/-	S7/-	S7/-	S7/-	S7/-

Table 2.2.7.4a View Prescription State Table

Test Coverage ID	Test Coverage
TCOV-07-018	S1 to S2 with input SP
TCOV-07-019	S2 to S3 with input PB
TCOV-07-020	S3 to S4 with input PS
TCOV-07-021	S3 to S5 with input PF
TCOV-07-022	S4 to S7 with input GP
TCOV-07-023	S1 to S6 with input NLI

Table 2.2.7.4b View Prescription State Coverage

2.2.7.5 Use Case Testing

Use Case ID	UC007
Use Case	F007 View Prescription

Purpose	To allow patients to view prescription details, make bill payment, and generate a bill in PDF format.
Actor	Patient
Trigger	Patient selects the Prescriptions function from the dashboard.
Precondition	Patient is logged into the system and has at least one prescription record available.
Main Flow	1. Patient selects Prescriptions. 2. System retrieves prescription records. 3. System displays prescription details including doctor name, appointment ID, appointment date, appointment time, disease, allergies, and medication. 4. Patient clicks Pay Bill. 5. System processes bill payment. 6. System displays "Bill Paid Successfully". 7. Patient clicks OK. 8. System generates and displays the bill in PDF format.
Alternate Flow – No Prescription	1.1 Patient has no prescription records. 1.2 System displays an empty list. 1.3 Back to Main Flow Step 1.
Alternate Flow – Payment Failed	2.1 Bill payment is not completed successfully. 2.2 System does not generate the PDF bill. 2.3 Back to Main Flow Step 4.
Rules	i. Patient must be logged in. ii. Only prescriptions belonging to the logged-in patient shall be displayed. iii. Bill must be paid before PDF bill generation.
Note	PDF bill generation is only allowed after successful bill payment.

Table 2.2.7.5a View Prescription Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-07-018	Main flow	TCOV-07-024	Prescription, payment, and PDF generated	Patient with prescription, payment success
TCON-07-019	Alternate Flow – No prescription	TCOV-07-025	Empty list displayed	No prescription records
TCON-07-020	Alternate Flow – Payment failed	TCOV-07-026	PDF not generated	Payment failed

Table 2.2.7.5b View Prescription Use Case Test Conditions & Coverage

2.2.8 F008 Submit Contact Query

2.2.8.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-08-001	All inputs are valid	TCOV-08-001	Query submission success	Name: John, Email: john@gmail.com, Phone: 0123456789, Message: Need help
TCON-08-002	Name field is empty	TCOV-08-002	Missing name validation	Name: empty
TCON-08-003	Email format is invalid	TCOV-08-003	Invalid email error	Email: johngmail.com
TCON-08-004	Phone number is empty	TCOV-08-004	Missing phone validation	Phone: empty
TCON-08-005	Phone number format is invalid	TCOV-08-005	Invalid phone error	Phone: abc123
TCON-08-006	Message field is empty	TCOV-08-006	Missing message validation	Message: empty
TCON-08-007	All fields are empty	TCOV-08-007	Validation error for all inputs	All fields empty

Table 2.2.8.1 Submit Contact Query Equivalence Partitioning Test Conditions & Coverage

2.2.8.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-08-008	Phone number length = 0	TCOV-08-008	Empty phone invalid	Phone: empty
TCON-08-009	Phone number contains alphabetic characters	TCOV-08-009	Invalid phone rejected	Phone: abc123
TCON-08-010	Phone number contains numeric characters	TCOV-08-010	Valid phone accepted	Phone: 0123456789

TCOV-08-011	Message length = 0 characters	TCOV-08-011	Empty message invalid	Message: empty
TCOV-08-012	Message length = 1 character	TCOV-08-012	Minimum message accepted	Message: H

Table 2.2.8.2 Submit Contact Query Boundary Value Analysis Test Conditions & Coverage

2.2.8.3 Decision Table Testing

Test Condition ID	Test Condition
TCOV-08-013	Name provided
TCOV-08-014	Email valid
TCOV-08-015	Phone number valid
TCOV-08-016	Message provided
TCOV-08-017	Missing field
TCOV-08-018	Invalid input

Table 2.2.8.3a Submit Contact Query Decision Table Test Conditions

Condition	TCOV-08-013	TCOV-08-014	TCOV-08-015	TCOV-08-016
Name Provided	Y	Y	Y	N
Email Valid	Y	N	Y	Y
Phone Valid	Y	Y	N	Y
Message Provided	Y	Y	Y	Y
Output				
Submit Query	Y	N	N	N
Show Validation/Error Message	N	Y	Y	Y

Table 2.2.8.3b Submit Contact Query Decision Table

2.2.8.4 State Transition Testing

State	Enter Details (ED)	Submit Query (SQ)	Valid Input (VI)	Invalid Input (II)
Contact Page (S1)	S2/Input Entered	S1/-	S1/-	S1/-
Input Entered (S2)	S2/-	S3/Validate	S2/-	S2/-
Validation (S3)	S3/-	S3/-	S4/Submit Success	S5/Error Message
Success Message (S4)	S4/-	S4/-	S4/-	S4/-

Error Displayed (S5)	S2/Re-enter Details	S5/-	S5/-	S5/-
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Table 2.2.8.4a Submit Contact Query State Table

Test Coverage ID	Test Coverage
TCOV-08-015	S1 to S2 with input ED
TCOV-08-016	S2 to S3 with input SQ
TCOV-08-017	S3 to S4 with input VI
TCOV-08-018	S3 to S5 with input II
TCOV-08-019	S5 to S2 with input ED

Table 2.2.8.4b Submit Contact Query State Coverage

2.2.8.5 Use Case Testing

Use Case ID	UC008
Use Case	F008 Submit Contact Query
Purpose	To allow users to send inquiries or feedback to the system administrator.
Actor	Patient
Trigger	Patient fills in the contact form and clicks submit.
Precondition	Contact form is accessible to the user.
Main Flow	1. Patient opens the Contact page. 2. System displays contact form. 3. Patient enters name, email, phone number, and message. 4. Patient clicks Submit button. 5. System validates the input. 6. System saves the query. 7. System displays success message.
Alternate Flow – Missing Field	1.1 Patient leaves one or more fields empty. 1.2 System displays validation message. 1.3 Back to Main Flow Step 3.
Alternate Flow – Invalid Email	2.1 Patient enters invalid email format. 2.2 System displays error message. 2.3 Back to Main Flow Step 3.
Rules	i. All fields must be filled. ii. Email must be valid format. iii. Phone number must be provided in valid contact number format. iv. Message must not be empty.
Note	Queries are stored for admin review.

Table 2.2.8.5a Submit Contact Query Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-08-019	Main flow	TCOV-08-020	Query submitted successfully	Name: John, Email: john@gmail.com, Phone: 0123456789, Message: Help needed
TCON-08-020	Alternate Flow – Missing field	TCOV-08-021	Validation error shown	Phone: empty
TCON-08-021	Alternate Flow – Invalid email	TCOV-08-022	Email error shown	Email: johngmail.com
TCON-08-022	Alternate Flow – Invalid phone number	TCOV-08-023	Phone validation error shown	Phone: abc123

Table 2.2.8.5b Submit Contact Query Use Case Test Conditions & Coverage

2.2.9 F009 View Appointment

2.2.9.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-09-001	Doctor is logged in and has appointments	TCOV-09-001	Appointment list displayed	Doctor with 3 appointments
TCON-09-002	Doctor is logged in but has no appointments	TCOV-09-002	Empty appointment list displayed	Doctor with 0 appointments
TCON-09-003	Search by valid patient contact number	TCOV-09-003	Matching appointment records displayed	Contact: 0123456789
TCON-09-004	Search by unmatched patient contact number	TCOV-09-004	No result message displayed	Contact: 0199999999
TCON-09-005	Doctor is not logged in	TCOV-09-005	Redirect to login	Direct URL access

Table 2.2.9.1 View Appointment Equivalence Partitioning Test Conditions & Coverage

2.2.9.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-09-006	Number of appointments = 0	TCOV-09-006	Empty list displayed	0 records
TCON-09-007	Number of appointments = 1	TCOV-09-007	Single appointment displayed	1 record
TCON-09-008	Number of appointments > 1	TCOV-09-008	Multiple appointments displayed	5 records
TCON-09-009	Appointment date = past date	TCOV-09-009	Past appointment displayed	Date: 2026-04-01
TCON-09-010	Appointment date = future date	TCOV-09-010	Upcoming appointment displayed	Date: 2026-05-01

Table 2.2.9.2 View Appointment Boundary Value Analysis Test Conditions & Coverage

2.2.9.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-09-011	Doctor is logged in
TCON-09-012	Appointment records exist
TCON-09-013	Appointment records do not exist
TCON-09-014	Appointment record valid
TCON-09-015	Appointment record invalid

Table 2.2.9.3a View Appointment Decision Table Test Conditions

Condition	TCOV-09-011	TCOV-09-012	TCOV-09-013	TCOV-09-014
Doctor Logged In	Y	Y	Y	N
Appointment Records Exist	Y	Y	N	-
Record Valid	Y	N	-	-
Output				
Display Appointment List	Y	N	N	N
Display Empty List	N	Y	Y	N
Show Error Message	N	N	N	Y
Redirect to Login Page	N	N	N	Y

Table 2.2.9.3b View Appointment Decision Table

2.2.9.4 State Transition Testing

State	Select View Appointment (SVA)	Records Found (RF)	No Records (NR)	Invalid Record (IR)	Not Logged In (NLI)
Doctor Dashboard (S1)	S2/Retrieve Appointments	S1/-	S1/-	S1/-	S5/Login Page
Retrieve Appointments (S2)	S2/-	S3/Display List	S4/Display Empty	S4/Error Message	S5/Login Page
Appointment List Displayed (S3)	S3/-	S3/-	S3/-	S3/-	S3/-
Empty/Error Display (S4)	S1/Back to Dashboard	S4/-	S4/-	S4/-	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.9.4a View Appointment State Table

Test Coverage ID	Test Coverage
TCOV-09-015	S1 to S2 with input SVA
TCOV-09-016	S2 to S3 with input RF
TCOV-09-017	S2 to S4 with input NR
TCOV-09-018	S2 to S4 with input IR
TCOV-09-019	S1 to S5 with input NLI

Table 2.2.9.4b View Appointment State Coverage

2.2.9.5 Use Case Testing

Use Case ID	UC009
Use Case	F009 View Appointment
Purpose	To allow doctors to view patient appointments.
Actor	Doctor
Trigger	Doctor selects the View Appointment function from the dashboard.
Precondition	Doctor is logged into the system.
Main Flow	1. Doctor selects View Appointment. 2. System retrieves appointment records. 3. System displays appointment list. 4. Doctor views details (patient name, date, time, status).

Alternate Flow – No Appointment	1.1 Doctor has no appointment records. 1.2 System displays an empty list or appropriate message. 1.3 Back to Main Flow Step 1.
Alternate Flow – Search by Contact Number	2.1 Doctor enters a patient contact number in the search field. 2.2 System filters appointment records based on the contact number. 2.3 System displays matching appointment records.
Alternate Flow – No Search Result	3.1 Search contact number does not match any appointment. 3.2 System displays an appropriate message and remains on the appointment list page.
Rules	i. Doctor must be logged in. ii. Only appointments assigned to the doctor shall be displayed. iii. Appointment data is read-only.
Note	Doctors cannot modify appointments in this function.

Table 2.2.9.5a View Appointment Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCOV-09-016	Main flow	TCOV-09-020	Appointment list displayed	Doctor with appointments
TCOV-09-017	Alternate Flow – No appointment	TCOV-09-021	Empty list displayed	No appointments
TCOV-09-018	Alternate Flow – Search by Contact Number	TCOV-09-022	Matching appointment records displayed	Contact: 0123456789
TCOV-09-019	Alternate Flow – No Search Result	TCOV-09-023	No result message displayed	Contact: 0199999999

Table 2.2.9.5b View Appointment Use Case Test Conditions & Coverage

2.2.10 F010 Prescribe Medication

2.2.10.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-10-001	Doctor is logged in, appointment is active, and prescription	TCOV-10-001	Prescription created successfully	Disease: Flu, Allergy: None, Prescription: Paracetamol

	details are valid			
TCON-10-002	Disease field is empty	TCOV-10-002	Missing disease validation	Disease: empty
TCON-10-003	Prescription detail field is empty	TCOV-10-003	Missing prescription validation	Prescription: empty
TCON-10-004	Selected appointment is invalid	TCOV-10-004	Prescription creation rejected	Appointment ID: invalid
TCON-10-005	Selected appointment is not active	TCOV-10-005	Prescription creation rejected	Appointment Status: Cancelled
TCON-10-006	Doctor is not logged in	TCOV-10-006	Access denied / redirect to login	Direct prescribe URL access

Table 2.2.10.1 Prescribe Medication Equivalence Partitioning Test Conditions & Coverage

2.2.10.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-10-007	Disease length = 0 characters	TCOV-10-007	Disease field invalid	Disease: empty
TCON-10-008	Disease length = 1 character	TCOV-10-008	Minimum disease text accepted	Disease: A
TCON-10-009	Prescription length = 0 characters	TCOV-10-009	Prescription field invalid	Prescription: empty
TCON-10-010	Prescription length = 1 character	TCOV-10-010	Minimum prescription text accepted	Prescription: X
TCON-10-011	Allergy field is empty	TCOV-10-011	Allergy field accepted as optional text	Allergy: empty
TCON-10-012	Allergy field contains text	TCOV-10-012	Allergy text accepted	Allergy: Penicillin

Table 2.2.10.2 Prescribe Medication Boundary Value Analysis Test Conditions & Coverage

2.2.10.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-10-013	Doctor is logged in

TCOV-10-014	Appointment exists
TCOV-10-015	Appointment is active
TCOV-10-016	Disease field filled
TCOV-10-017	Prescription field filled
TCOV-10-018	Required field missing
TCOV-10-019	Appointment invalid or inactive

Table 2.2.10.3a Prescribe Medication Decision Table Test Conditions

Condition	TCOV-10-013	TCOV-10-014	TCOV-10-015	TCOV-10-016
Doctor Logged In	Y	Y	Y	N
Appointment Exists	Y	Y	N	-
Appointment Active	Y	Y	N	-
Disease Filled	Y	N	Y	-
Prescription Filled	Y	Y	N	-
Output				
Create Prescription Record	Y	N	N	N
Display Success Message	Y	N	N	N
Display Validation Message	N	Y	N	N
Reject Prescription Creation	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.10.3b Prescribe Medication Decision Table

2.2.10.4 State Transition Testing

State	Select Appointment (SA)	Click Prescribe (CP)	Enter Details (ED)	Submit Prescription (SP)	Valid Input (VI)	Invalid Input (II)	Invalid Appointment (IA)
Appointment List (S1)	S2/Select Appointment	S1/-	S1/-	S1/-	S1/-	S1/-	S5/Error Message
Appointment Selected (S2)	S2/-	S3/Display Prescription Form	S2/-	S2/-	S2/-	S2/-	S5/Error Message

Prescription Form (S3)	S3/-	S3/-	S4/Input Entered	S3/-	S3/-	S3/-	S5/Error Message
Validation (S4)	S4/-	S4/-	S4/-	S4/Validate	S6/Save Prescription	S5/Validation Message	S5/Error Message
Error Displayed (S5)	S1/Back to Appointment List	S5/-	S3/Re-enter Details	S5/-	S5/-	S5/-	S5/-
Prescription Created (S6)	S6/-	S6/-	S6/-	S6/-	S7/Success Message	S6/-	S6/-
Success Displayed (S7)	S1/Back to Appointment List	S7/-	S7/-	S7/-	S7/-	S7/-	S7/-

Table 2.2.10.4a Prescribe Medication State Table

Test Coverage ID	Test Coverage
TCOV-10-017	S1 to S2 with input SA
TCOV-10-018	S2 to S3 with input CP
TCOV-10-019	S3 to S4 with input ED
TCOV-10-020	S4 to S6 with input VI
TCOV-10-021	S4 to S5 with input II
TCOV-10-022	S1/S2 to S5 with input IA
TCOV-10-023	S6 to S7 with success message
TCOV-10-024	S7 to S1 with back to appointment list

Table 2.2.10.4b Prescribe Medication State Coverage

2.2.10.5 Use Case Testing

Use Case ID	UC010
Use Case	F010 Prescribe Medication
Purpose	To allow doctors to create a prescription for a selected patient appointment.
Actor	Doctor
Trigger	Doctor clicks the Prescribe action for a selected appointment.

Precondition	Doctor is logged into the system, the selected appointment exists, and the appointment is active.
Main Flow	1. Doctor opens the Appointment page. 2. System displays the list of appointments assigned to the doctor. 3. Doctor selects the Prescribe action for a chosen appointment. 4. System displays the prescription entry form. 5. Doctor enters disease, allergy, and prescription details. 6. Doctor clicks the Submit button. 7. System validates the entered prescription information. 8. System stores the prescription record linked to the selected appointment and patient. 9. System displays a success message. 10. System redirects the doctor to the relevant doctor page.
Alternate Flow – Missing Required Field	1.1 Doctor leaves disease or prescription field empty. 1.2 System displays a validation message. 1.3 System does not save the prescription. 1.4 Back to Main Flow Step 5.
Alternate Flow – Invalid or Inactive Appointment	2.1 Selected appointment is invalid or no longer active. 2.2 System does not create the prescription. 2.3 System returns to the appointment list.
Alternate Flow – Save Failed	3.1 Prescription record cannot be stored successfully. 3.2 System displays an error message. 3.3 No prescription record is created.
Rules	i. Prescription shall be linked to a selected appointment record. ii. Only the logged-in doctor can create prescriptions for appointments assigned to that doctor. iii. Disease and prescription details must be entered before submission.
Note	Allergy information may be recorded if available.

Table 2.2.10.5a Prescribe Medication Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCO-10-020	Main flow	TCOV-10-025	Prescription created successfully	Disease: Flu, Allergy: None, Prescription: Paracetamol 500mg

TCON-10-021	Alternate Flow – Missing required field	TCOV-10-026	Validation message displayed	Disease: empty, Prescription: Paracetamol
TCON-10-022	Alternate Flow – Invalid or inactive appointment	TCOV-10-027	Prescription creation rejected	Appointment Status: Cancelled
TCON-10-023	Alternate Flow – Save failed	TCOV-10-028	Error message displayed and record not saved	Database insert failed

Table 2.2.10.5b Prescribe Medication Use Case Test Conditions & Coverage

2.2.11 F011 Cancel Appointment

2.2.11.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-11-001	Doctor is logged in and appointment is active	TCOV-11-001	Cancellation success	Logged-in doctor, Status: Active
TCON-11-002	Doctor is logged in but appointment already cancelled	TCOV-11-002	Cancellation rejected	Status: Cancelled
TCON-11-003	Doctor confirms cancellation	TCOV-11-003	Status updated to cancelled	Click OK
TCON-11-004	Doctor rejects cancellation	TCOV-11-004	Appointment remains unchanged	Click Cancel
TCON-11-005	Appointment not assigned to doctor	TCOV-11-005	Cancellation not allowed	Different doctor's appointment
TCON-11-006	Doctor not logged in	TCOV-11-006	Access denied / redirect to login	Direct URL access

Table 2.2.11.1 Cancel Appointment Equivalence Partitioning Test Conditions & Coverage

2.2.11.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
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TCON-11-007	Number of active appointments = 0	TCOV-11-007	No cancel action available	0 active appointments
TCON-11-008	Number of active appointments = 1	TCOV-11-008	Single cancel action available	1 active appointment
TCON-11-009	Number of active appointments > 1	TCOV-11-009	Multiple cancel actions available	4 active appointments
TCON-11-010	Status transition Active → Cancelled	TCOV-11-010	Status update boundary	Active → Cancelled

Table 2.2.11.2 Cancel Appointment Boundary Value Analysis Test Conditions & Coverage

2.2.11.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-11-011	Doctor logged in
TCON-11-012	Appointment exists
TCON-11-013	Appointment is active
TCON-11-014	Appointment assigned to doctor
TCON-11-015	Confirmation accepted
TCON-11-016	Confirmation rejected

Table 2.2.11.3a Cancel Appointment Decision Table Test Conditions

Condition	TCOV-11-011	TCOV-11-012	TCOV-11-013	TCOV-11-014	TCOV-11-015
Doctor Logged In	Y	Y	Y	Y	N
Appointment Exists	Y	Y	Y	Y	-
Appointment Active	Y	Y	N	Y	-
Assigned to Doctor	Y	Y	Y	N	-
Confirm Cancel	Y	N	-	-	-
Output					
Cancel Appointment	Y	N	N	N	N

Show Success Message	Y	N	N	N	N
Keep Appointment	N	Y	N	N	Y
Show Error Message	N	N	Y	Y	N
Redirect to Login	N	N	N	N	Y

Table 2.2.11.3b Cancel Appointment Decision Table

2.2.11.4 State Transition Testing

State	Select Appointment (SA)	Click Cancel (CC)	Confirm Cancel (CF)	Reject Cancel (RC)	Not Assigned (NA)	Not Logged In (NLI)
Doctor Dashboard (S1)	S2/View Appointments	S1/-	S1/-	S1/-	S1/-	S6/Login Page
Appointment List (S2)	S2/-	S3/Confirm Dialog	S2/-	S2/-	S5/Error Message	S6/Login Page
Confirmation Dialog (S3)	S3/-	S3/-	S4/Update Status	S2/Back	S3/-	S6/Login Page
Appointment Cancelled (S4)	S4/-	S4/-	S4/-	S4/-	S4/-	S4/-
Error Displayed (S5)	S2/Back	S5/-	S5/-	S5/-	S5/-	S6/Login Page
Login Page (S6)	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-

Table 2.2.11.4a Cancel Appointment State Table

Test Coverage ID	Test Coverage
TCOV-11-016	S1 to S2 with input SA
TCOV-11-017	S2 to S3 with input CC
TCOV-11-018	S3 to S4 with input CF
TCOV-11-019	S3 to S2 with input RC
TCOV-11-020	S2 to S5 with input NA
TCOV-11-021	S1 to S6 with input NLI

Table 2.2.11.4b Cancel Appointment State Coverage

2.2.11.5 Use Case Testing

Use Case ID	UC011
Use Case	F011 Cancel Appointment

Purpose	To allow doctors to cancel an assigned appointment.
Actor	Doctor
Trigger	Doctor clicks Cancel on an appointment.
Precondition	Doctor is logged in and appointment belongs to doctor.
Main Flow	1. Doctor opens appointment list. 2. System displays appointments. 3. Doctor selects Cancel. 4. System shows confirmation dialog. 5. Doctor confirms. 6. System updates status to Cancelled. 7. System displays success message. 8. Return to dashboard.
Alternate Flow – Reject Confirmation	1.1 Doctor clicks Cancel in dialog. 1.2 System keeps appointment unchanged. 1.3 Return to appointment list.
Alternate Flow – Not Assigned Appointment	2.1 Doctor selects appointment not assigned. 2.2 System blocks action. 2.3 Show error message.
Alternate Flow – Not Logged In	3.1 User tries access without login. 3.2 Redirect to login page.
Rules	i. Only assigned doctor can cancel. ii. Only active appointments can be cancelled. iii. Confirmation required before cancel.
Note	Cancellation updates status only (no deletion).

Table 2.2.11.5a Cancel Appointment Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-11-017	Main flow	TCOV-11-022	Appointment cancelled successfully	Status: Active, Confirm: OK
TCON-11-018	Alternate Flow – Reject confirmation	TCOV-11-023	Cancellation aborted	Confirm: Cancel
TCON-11-019	Alternate Flow – Not assigned	TCOV-11-024	Action blocked	Different doctor's appointment
TCON-11-020	Alternate Flow – Not logged in	TCOV-11-025	Redirect to login	Direct access without login

Table 2.2.11.5b Cancel Appointment Use Case Test Conditions & Coverage

2.2.12 F012 View Prescription List

2.2.12.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-12-001	Doctor is logged in and prescription records exist	TCOV-12-001	Prescription list displayed	Doctor with 3 prescription records
TCON-12-002	Doctor is logged in but no prescription records exist	TCOV-12-002	Empty prescription list displayed	Doctor with 0 prescription records
TCON-12-003	Doctor is not logged in	TCOV-12-003	Access denied / redirect to login	Direct URL access
TCON-12-004	Prescription record is valid	TCOV-12-004	Prescription details displayed correctly	Disease: Flu, Allergy: None, Prescription: Paracetamol
TCON-12-005	Prescription record is invalid/missing	TCOV-12-005	Error or empty result handled	Invalid prescription ID

Table 2.2.12.1 View Prescription List Equivalence Partitioning Test Conditions & Coverage

2.2.12.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-12-006	Number of prescription records = 0	TCOV-12-006	Empty list displayed	0 records
TCON-12-007	Number of prescription records = 1	TCOV-12-007	Single prescription displayed	1 record
TCON-12-008	Number of prescription records > 1	TCOV-12-008	Multiple prescriptions displayed	5 records
TCON-12-009	Prescription linked to logged-in doctor	TCOV-12-009	Record displayed	Doctor ID matches logged-in doctor
TCON-12-010	Prescription not linked to logged-in doctor	TCOV-12-010	Record not displayed	Different doctor's prescription

Table 2.2.12.2 View Prescription List Boundary Value Analysis Test Conditions & Coverage

2.2.12.3 Decision Table Testing

Test Condition ID	Test Condition
TCOV-12-011	Doctor is logged in
TCOV-12-012	Prescription records exist
TCOV-12-013	Prescription records do not exist
TCOV-12-014	Prescription belongs to logged-in doctor
TCOV-12-015	Prescription does not belong to logged-in doctor

Table 2.2.12.3a View Prescription List Decision Table Test Conditions

Condition	TCOV-12-011	TCOV-12-012	TCOV-12-013	TCOV-12-014
Doctor Logged In	Y	Y	Y	N
Prescription Records Exist	Y	Y	N	-
Belongs to Logged-in Doctor	Y	N	-	-
Output				
Display Prescription List	Y	N	N	N
Hide Unauthorized Records	N	Y	N	N
Display Empty List / Message	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.12.3b View Prescription List Decision Table

2.2.12.4 State Transition Testing

State	Select Prescription List (SPL)	Records Found (RF)	No Records (NR)	Unauthorized Record (UR)	Not Logged In (NLI)
Doctor Dashboard (S1)	S2/Retrieve Prescriptions	S1/-	S1/-	S1/-	S5/Login Page
Retrieve Prescription Records (S2)	S2/-	S3/Display List	S4/Display Empty List	S4/Hide Record	S5/Login Page
Prescription List	S3/-	S3/-	S3/-	S3/-	S3/-

Displayed (S3)					
Empty/Error Displayed (S4)	S1/Back to Dashboard	S4/-	S4/-	S4/-	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.12.4a View Prescription List State Table

Test Coverage ID	Test Coverage
TCOV-12-015	S1 to S2 with input SPL
TCOV-12-016	S2 to S3 with input RF
TCOV-12-017	S2 to S4 with input NR
TCOV-12-018	S2 to S4 with input UR
TCOV-12-019	S1 to S5 with input NLI

Table 2.2.12.4b View Prescription List State Coverage

2.2.12.5 Use Case Testing

Use Case ID	UC012
Use Case	F012 View Prescription List
Purpose	To allow doctors to review prescription records associated with their patients.
Actor	Doctor
Trigger	Doctor selects the Prescription List function from the doctor dashboard.
Precondition	Doctor is logged into the system and the prescription list page is available.
Main Flow	1. Doctor selects the Prescription List function. 2. System retrieves prescription records associated with patients assigned to the logged-in doctor. 3. System displays the prescription list, including appointment details, disease, allergy, and prescription information. 4. Doctor reviews the prescription records.
Alternate Flow – No Prescription Records	1.1 No prescription records are associated with the doctor's patients. 1.2 System displays an empty list or appropriate message. 1.3 Back to Main Flow Step 1.
Alternate Flow – Unauthorized Record	2.1 Prescription record does not belong to the logged-in doctor's patients. 2.2 System does not display the unauthorized record.
Alternate Flow – Not Logged In	3.1 User attempts to access the prescription list without login. 3.2 System redirects user to the login page.
Rules	i. Only prescription records associated with the logged-in doctor's patients shall be displayed.

	ii. Prescription information in this view shall be displayed in read-only form.
Note	Doctor can view prescription records but cannot edit them from this view.

Table 2.2.12.5a View Prescription List Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-12-016	Main flow	TCOV-12-020	Prescription list displayed	Doctor with existing prescription records
TCON-12-017	Alternate Flow – No prescription records	TCOV-12-021	Empty prescription list displayed	Doctor with 0 prescription records
TCON-12-018	Alternate Flow – Unauthorized record	TCOV-12-022	Unauthorized prescription hidden	Prescription belongs to another doctor
TCON-12-019	Alternate Flow – Not logged in	TCOV-12-023	Redirect to login page	Direct URL access without login

Table 2.2.12.5b View Prescription List Use Case Test Conditions & Coverage

2.2.13 F013 Add Doctor

2.2.13.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-13-001	Admin is logged in and all doctor details are valid	TCOV-13-001	Doctor account created successfully	Name: Dr John, Specialization: Cardiologist, Email: doctor@gmail.com, Password: doc123, Confirm: doc123, Fee: 100
TCON-13-002	Required field is empty	TCOV-13-002	Missing input validation	Doctor Name: empty
TCON-13-003	Email format is invalid	TCOV-13-003	Invalid email rejected	Email: doctorgmail.com
TCON-13-004	Password and confirm password do not match	TCOV-13-004	Password mismatch rejected	Password: doc123, Confirm: doc124

TCON-13-005	Consultation fee is invalid	TCOV-13-005	Invalid fee rejected	Fee: -50
TCON-13-006	Admin is not logged in	TCOV-13-006	Access denied / redirect to login	Direct add doctor URL access

Table 2.2.13.1 Add Doctor Equivalence Partitioning Test Conditions & Coverage

2.2.13.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-13-007	Password length = 0 characters	TCOV-13-007	Empty password invalid	Password: empty
TCON-13-008	Password length = 5 characters	TCOV-13-008	Password below minimum invalid	Password: doc12
TCON-13-009	Password length = 6 characters	TCOV-13-009	Minimum password length valid	Password: doc123
TCON-13-010	Password length = 7 characters	TCOV-13-010	Password above minimum valid	Password: doc1234
TCON-13-011	Consultation fee = 0	TCOV-13-011	Zero fee invalid	Fee: 0
TCON-13-012	Consultation fee > 0	TCOV-13-012	Positive fee valid	Fee: 100

Table 2.2.13.2 Add Doctor Boundary Value Analysis Test Conditions & Coverage

2.2.13.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-13-013	Admin is logged in
TCON-13-014	All required fields are filled
TCON-13-015	Email is valid
TCON-13-016	Password matches confirm password
TCON-13-017	Consultation fee is positive
TCON-13-018	Required field missing
TCON-13-019	Admin is not logged in

Table 2.2.13.3a Add Doctor Decision Table Test Conditions

Condition	TCOV-13-013	TCOV-13-014	TCOV-13-015	TCOV-13-016	TCOV-13-017
Admin Logged In	Y	Y	Y	Y	N
All Required Fields Filled	Y	N	Y	Y	-
Email Valid	Y	Y	N	Y	-
Password Match	Y	Y	Y	N	-
Fee Positive	Y	Y	Y	Y	-
Output					
Create Doctor Record	Y	N	N	N	N
Display “Doctor added successfully!”	Y	N	N	N	N
Display Validation/Error Message	N	Y	Y	Y	N
Redirect to Login Page	N	N	N	N	Y

Table 2.2.13.3b Add Doctor Decision Table

2.2.13.4 State Transition Testing

State	Select Add Doctor (SAD)	Enter Details (ED)	Submit Form (SF)	Valid Input (VI)	Invalid Input (II)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/Display Add Doctor Form	S1/-	S1/-	S1/-	S1/-	S6/Login Page
Add Doctor Form (S2)	S2/-	S3/Input Entered	S2/-	S2/-	S2/-	S6/Login Page
Input Entered (S3)	S3/-	S3/-	S4/Validate	S3/-	S3/-	S6/Login Page
Validation (S4)	S4/-	S4/-	S4/-	S5/Create Doctor Record	S2/Error Message	S6/Login Page
Doctor Added (S5)	S1/Back to Dashboard	S5/-	S5/-	S5/-	S5/-	S5/-
Login Page (S6)	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-

Table 2.2.13.4a Add Doctor State Table

Test Coverage ID	Test Coverage
TCOV-13-018	S1 to S2 with input SAD
TCOV-13-019	S2 to S3 with input ED
TCOV-13-020	S3 to S4 with input SF
TCOV-13-021	S4 to S5 with input VI
TCOV-13-022	S4 to S2 with input II
TCOV-13-023	S1 to S6 with input NLI
TCOV-13-024	S5 to S1 with back to dashboard

Table 2.2.13.4b Add Doctor State Coverage

2.2.13.5 Use Case Testing

Use Case ID	UC013
Use Case	F013 Add Doctor
Purpose	To allow an admin to register a new doctor into the hospital management system.
Actor	Admin
Trigger	Admin navigates to the Add Doctor section and submits the required doctor details.
Precondition	Admin is successfully logged into the dashboard.
Main Flow	1. Admin selects the Add Doctor tab. 2. System displays the Add Doctor form. 3. Admin enters doctor name, specialization, email, password, confirm password, and consultancy fees. 4. Admin clicks Add Doctor. 5. System validates the input. 6. System creates a new doctor record. 7. System displays "Doctor added successfully!".
Alternate Flow – Missing Required Field	1.1 Admin leaves one or more required fields empty. 1.2 System displays a validation message. 1.3 Back to Main Flow Step 3.
Alternate Flow – Password Mismatch	2.1 Admin enters password and confirm password that do not match. 2.2 System displays "Not Matching". 2.3 Back to Main Flow Step 3.
Alternate Flow – Invalid Input	3.1 Admin enters invalid doctor details such as invalid email, invalid fee, or non-alphabetic name. 3.2 System rejects the input or database insertion fails. 3.3 System displays an error message.
Alternate Flow – Not Logged In	4.1 User attempts to access Add Doctor page without admin login. 4.2 System redirects user to the login page.
Rules	i. Admin must be logged in.

	ii. All fields are mandatory. iii. Password and confirm password must match. iv. Doctor name should contain alphabetic characters only.v. Consultation fee must be a positive numeric value.
Note	Added doctor account can be used for doctor login after successful creation.

Table 2.2.13.5a Add Doctor Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-13-020	Main flow	TCOV-13-025	Doctor added successfully	Name: Dr John, Specialization: Cardiologist, Email: doctor@gmail.com, Password: doc123, Confirm: doc123, Fee: 100
TCON-13-021	Alternate Flow – Missing required field	TCOV-13-026	Validation message displayed	Doctor Name: empty
TCON-13-022	Alternate Flow – Password mismatch	TCOV-13-027	Password mismatch handled	Password: doc123, Confirm: doc124
TCON-13-023	Alternate Flow – Invalid input	TCOV-13-028	Invalid input rejected	Email: doctorgmail.com, Fee: -50
TCON-13-024	Alternate Flow – Not logged in	TCOV-13-029	Redirect to login page	Direct add doctor URL access without login

Table 2.2.13.5b Add Doctor Use Case Test Conditions & Coverage

2.2.14 F014 Delete Doctor

2.2.14.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-14-001	Admin is logged in and doctor email exists	TCOV-14-001	Doctor deleted successfully	Email: doctor@gmail.com
TCON-14-002	Doctor email does not exist	TCOV-14-002	Deletion rejected	Email: unknown@gmail.com

TCON-14-003	Admin confirms deletion	TCOV-14-003	Doctor record removed	Confirm: OK
TCON-14-004	Admin denies confirmation	TCOV-14-004	Deletion aborted	Confirm: Cancel
TCON-14-005	Deletion query fails	TCOV-14-005	"Unable to delete!" displayed	Database delete failed
TCON-14-006	Admin is not logged in	TCOV-14-006	Redirect to login	Direct delete URL access

Table 2.2.14.1 Delete Doctor Equivalence Partitioning Test Conditions & Coverage

2.2.14.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-14-007	Number of doctors = 0	TCOV-14-007	No deletion possible	0 doctor records
TCON-14-008	Number of doctors = 1	TCOV-14-008	Single deletion available	1 doctor record
TCON-14-009	Number of doctors > 1	TCOV-14-009	Multiple doctor records available	5 doctor records
TCON-14-010	Existing doctor email entered	TCOV-14-010	Existing doctor email accepted	doctor@gmail.com
TCON-14-011	Non-existing doctor email entered	TCOV-14-011	Non-existing doctor email rejected	unknown@gmail.com

Table 2.2.14.2 Delete Doctor Boundary Value Analysis Test Conditions & Coverage

2.2.14.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-14-012	Admin is logged in
TCON-14-013	Doctor email exists
TCON-14-014	Confirmation accepted
TCON-14-015	Confirmation denied
TCON-14-016	Deletion query successful
TCON-14-017	Deletion query failed

Table 2.2.14.3a Delete Doctor Decision Table Test Conditions

Condition	TCOV-14-012	TCOV-14-013	TCOV-14-014	TCOV-14-015
Admin Logged In	Y	Y	Y	N
Doctor Email Exists	Y	Y	N	-
Confirmation Accepted	Y	N	-	-
Deletion Query Successful	Y	-	-	-
Output				
Delete Doctor Record	Y	N	N	N
Display “Doctor removed successfully!”	Y	N	N	N
Abort Deletion	N	Y	N	N
Display “Unable to delete!” / Error	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.14.3b Delete Doctor Decision Table

2.2.14.4 State Transition Testing

State	Open Delete Doctor Page (ODD)	Enter Email ID (EE)	Submit Delete (SD)	Confirm Delete (CD)	Reject Delete (RD)	Delete Failed (DF)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/Display Delete Form	S1/-	S1/-	S1/-	S1/-	S1/-	S6/Login Page
Delete Doctor Form (S2)	S2/-	S3/Input Entered	S2/-	S2/-	S2/-	S2/-	S6/Login Page
Email Entered (S3)	S3/-	S3/-	S4/Confirmation Dialog	S3/-	S3/-	S3/-	S6/Login Page
Confirmation Dialog (S4)	S4/-	S4/-	S4/-	S5/Delete Record	S2/Back to Form	S7/Error Message	S6/Login Page
Doctor Deleted (S5)	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-
Login Page (S6)	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-	S6/-
Error Displayed (S7)	S2/Back to Form	S7/-	S7/-	S7/-	S7/-	S7/-	S6/Login Page

Table 2.2.14.4a Delete Doctor State Table

Test Coverage ID	Test Coverage
TCOV-14-018	S1 to S2 with input ODD
TCOV-14-019	S2 to S3 with input EE
TCOV-14-020	S3 to S4 with input SD
TCOV-14-021	S4 to S5 with input CD
TCOV-14-022	S4 to S2 with input RD
TCOV-14-023	S4 to S7 with input DF
TCOV-14-024	S1 to S6 with input NLI

Table 2.2.14.4b Delete Doctor State Coverage

2.2.14.5 Use Case Testing

Use Case ID	UC014
Use Case	F014 Delete Doctor
Purpose	To remove a doctor's access and records from the active hospital management system.
Actor	Admin
Trigger	Admin navigates to the Delete Doctor section and submits the doctor's Email ID.
Precondition	Admin is logged in and the target doctor's email exists in the database.
Main Flow	- Admin selects the Delete Doctor tab. - System displays the Delete Doctor form. - Admin enters the target doctor Email ID. - Admin clicks Delete Doctor. - System prompts for confirmation. - Admin confirms the deletion. - System deletes the doctor record. - System displays "Doctor removed successfully!".
Alternate Flow – Confirmation Denied	1.1 Admin denies the confirmation prompt. 1.2 System aborts the deletion. 1.3 Back to Main Flow Step 3.
Alternate Flow – Delete Failed	2.1 Deletion query fails. 2.2 System displays "Unable to delete!". 2.3 Back to Main Flow Step 3.
Alternate Flow – Not Logged In	3.1 User attempts to access Delete Doctor without admin login. 3.2 System redirects user to login page.
Rules	- Only admin can delete doctor. - The Email ID must belong to an existing doctor in doctb. - Confirmation is required before deletion.
Note	Doctor deletion is performed using registered Email ID.

Table 2.2.14.5a Delete Doctor Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-14-018	Main flow	TCOV-14-025	Doctor deleted successfully	Email: doctor@gmail.com, Confirm: OK
TCON-14-019	Alternate Flow – Confirmation denied	TCOV-14-026	Deletion cancelled	Confirm: Cancel
TCON-14-020	Alternate Flow – Delete failed	TCOV-14-027	Error message displayed	Database delete failed
TCON-14-021	Alternate Flow – Not logged in	TCOV-14-028	Redirect to login	Direct delete URL access without login

Table 2.2.14.5b Delete Doctor Use Case Test Conditions & Coverage

2.2.15 F015 View & Search Doctor List

2.2.15.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-15-001	Admin is logged in and doctor records exist	TCOV-15-001	Doctor list displayed	5 doctor records
TCON-15-002	Admin is logged in but no doctor records exist	TCOV-15-002	Empty list displayed	0 records
TCON-15-003	Admin is not logged in	TCOV-15-003	Access denied / redirect to login	Direct URL access
TCON-15-004	Search keyword matches doctor Email ID	TCOV-15-004	Matching doctor record displayed	Email: doctor@gmail.com
TCON-15-005	Search keyword does not match any doctor Email ID	TCOV-15-005	No results message displayed	Email: unknown@gmail.com
TCON-15-006	Search input is empty	TCOV-15-006	Full doctor list displayed	Empty search

Table 2.2.15.1 View & Search Doctor List Equivalence Partitioning Test Conditions & Coverage

2.2.15.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-15-007	Number of doctor records = 0	TCOV-15-007	Empty list displayed	0 records
TCON-15-008	Number of doctor records = 1	TCOV-15-008	Single record displayed	1 record
TCON-15-009	Number of doctor records > 1	TCOV-15-009	Multiple records displayed	10 records
TCON-15-010	Search Email ID length = 0	TCOV-15-010	Show all doctor records	Empty keyword
TCON-15-011	Search Email ID is valid format	TCOV-15-011	Search input accepted	doctor@gmail.com
TCON-15-012	Search Email ID is invalid format	TCOV-15-012	Search input rejected / no result shown	doctorgmail.com

Table 2.2.15.2 View & Search Doctor List Boundary Value Analysis Test Conditions & Coverage

2.2.15.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-15-013	Admin is logged in
TCON-15-014	Doctor records exist
TCON-15-015	Doctor Email ID provided
TCON-15-016	Search result found
TCON-15-017	Search result not found

Table 2.2.15.3a View & Search Doctor List Decision Table Test Conditions

Condition	TCOV-15-013	TCOV-15-014	TCOV-15-015	TCOV-15-016
Admin Logged In	Y	Y	Y	N
Doctor Records Exist	Y	Y	Y	-
Doctor Email ID Provided	N	Y	Y	-

Result Found	-	Y	N	-
Output				
Display Full Doctor List	Y	N	N	N
Display Search Results	N	Y	N	N
Display “No Results Found”	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.15.3b View & Search Doctor List Decision Table

2.2.15.4 State Transition Testing

State	Open Doctor List (ODL)	Enter Email ID (EE)	Submit Search (SS)	Results Found (RF)	No Results (NR)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/Display List	S1/-	S1/-	S1/-	S1/-	S5/Login Page
Doctor List Page (S2)	S2/-	S3/Input Entered	S2/-	S2/-	S2/-	S5/Login Page
Search Input (S3)	S3/-	S3/-	S4/Process Search	S3/-	S3/-	S5/Login Page
Search Results (S4)	S4/-	S3/New Search	S4/-	S4/Display Results	S4/Display No Results	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.15.4a View & Search Doctor List State Table

Test Coverage ID	Test Coverage
TCOV-15-017	S1 to S2 with input ODL
TCOV-15-018	S2 to S3 with input EE
TCOV-15-019	S3 to S4 with input SS
TCOV-15-020	S4 to S4 with input RF
TCOV-15-021	S4 to S4 with input NR
TCOV-15-022	S1 to S5 with input NLI

Table 2.2.15.4b View & Search Doctor List State Coverage

2.2.15.5 Use Case Testing

Use Case ID	UC015
Use Case	F015 View & Search Doctor

Purpose	To allow admin to view all registered doctors and search for a specific doctor by Email ID.
Actor	Admin
Trigger	Admin selects Doctor List or submits a doctor Email ID search query.
Precondition	Admin is logged in.
Main Flow	1. Admin opens doctor list page. 2. System displays all doctor records. 3. Admin enters a doctor Email ID into the search bar. 4. Admin clicks Search. 5. System filters doctor records based on the Email ID. 6. System displays the matching doctor details.
Alternate Flow – No Records	1.1 No doctor records exist. 1.2 System displays empty list. 1.3 Back to Main Flow Step 1.
Alternate Flow – No Search Results	2.1 The entered Email ID does not match any doctor record. 2.2 System displays “No results found”. 2.3 Back to Main Flow Step 3.
Alternate Flow – Not Logged In	3.1 User tries to access doctor list without login. 3.2 System redirects user to login page.
Rules	i. Admin must be logged in. ii. Search must be performed using doctor Email ID only. iii. Search results must be filtered correctly.
Note	Search is strictly based on the doctor’s registered Email ID.

Table 2.2.15.5a View & Search Doctor List Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-15-018	Main flow	TCOV-15-023	Doctor list and search results displayed	Email: doctor@gmail.com
TCON-15-019	Alternate Flow – No records	TCOV-15-024	Empty list displayed	0 records
TCON-15-020	Alternate Flow – No results	TCOV-15-025	No results message	Email: unknown@gmail.com
TCON-15-021	Alternate Flow – Not logged in	TCOV-15-026	Redirect to login	Direct access without login

Table 2.2.15.5b View & Search Doctor List Use Case Test Conditions & Coverage

2.2.16 F016 View & Search Patient List

2.2.16.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-16-001	Admin is logged in and patient records exist	TCOV-16-001	Patient list displayed	5 patient records
TCON-16-002	Admin is logged in but no patient records exist	TCOV-16-002	Empty list displayed	0 records
TCON-16-003	Admin is not logged in	TCOV-16-003	Access denied / redirect to login	Direct URL access
TCON-16-004	Search keyword matches patient contact number	TCOV-16-004	Matching patient record displayed	Contact: 0123456789
TCON-16-005	Search keyword does not match any patient contact number	TCOV-16-005	No results message displayed	Contact: 0199999999
TCON-16-006	Search input is empty	TCOV-16-006	Full patient list displayed	Empty search

Table 2.2.16.1 View & Search Patient List Equivalence Partitioning Test Conditions & Coverage

2.2.16.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-16-007	Number of patient records = 0	TCOV-16-007	Empty list displayed	0 records
TCON-16-008	Number of patient records = 1	TCOV-16-008	Single record displayed	1 record
TCON-16-009	Number of patient records > 1	TCOV-16-009	Multiple records displayed	10 records
TCON-16-010	Contact number length = 0	TCOV-16-010	Show all patient records	Empty keyword

TCOV-16-011	Contact number contains numeric characters	TCOV-16-011	Contact search accepted	0123456789
TCOV-16-012	Contact number contains alphabetic characters	TCOV-16-012	Invalid contact search rejected / no result shown	abc123

Table 2.2.16.2 View & Search Patient List Boundary Value Analysis Test Conditions & Coverage

2.2.16.3 Decision Table Testing

Test Condition ID	Test Condition
TCOV-16-013	Admin is logged in
TCOV-16-014	Patient records exist
TCOV-16-015	Contact number provided
TCOV-16-016	Search result found
TCOV-16-017	Search result not found

Table 2.2.16.3a View & Search Patient List Decision Table Test Conditions

Condition	TCOV-16-013	TCOV-16-014	TCOV-16-015	TCOV-16-016
Admin Logged In	Y	Y	Y	N
Patient Records Exist	Y	Y	Y	-
Contact Number Provided	N	Y	Y	-
Result Found	-	Y	N	-
Output				
Display Full Patient List	Y	N	N	N
Display Search Results	N	Y	N	N
Display "No Results Found"	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.16.3b View & Search Patient List Decision Table

2.2.16.4 State Transition Testing

State	Open Patient List (OPL)	Enter Contact Number (ECN)	Submit Search (SS)	Results Found (RF)	No Results (NR)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/Display Patient List	S1/-	S1/-	S1/-	S1/-	S5/Login Page
Patient List Page (S2)	S2/-	S3/Input Entered	S2/-	S2/-	S2/-	S5/Login Page
Search Input (S3)	S3/-	S3/-	S4/Process Search	S3/-	S3/-	S5/Login Page
Search Results (S4)	S4/-	S3/New Search	S4/-	S4/Display Results	S4/Display No Results	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.16.4a View & Search Patient List State Table

Test Coverage ID	Test Coverage
TCOV-16-017	S1 to S2 with input OPL
TCOV-16-018	S2 to S3 with input ECN
TCOV-16-019	S3 to S4 with input SS
TCOV-16-020	S4 to S4 with input RF
TCOV-16-021	S4 to S4 with input NR
TCOV-16-022	S1 to S5 with input NLI

Table 2.2.16.4b View & Search Patient List State Coverage

2.2.16.5 Use Case Testing

Use Case ID	UC016
Use Case	F016 View & Search Patient
Purpose	To allow admin to view all registered patients and search for a specific patient by contact number.
Actor	Admin
Trigger	Admin selects Patient List or submits a patient contact number search query.
Precondition	Admin is logged in.
Main Flow	1. Admin opens patient list page. 2. System displays all patient records. 3. Admin enters a patient contact number into the search bar.

	4. Admin clicks Search. 5. System filters patient records based on the contact number. 6. System displays the matching patient details.
Alternate Flow – No Records	1.1 No patient records exist. 1.2 System displays empty list. 1.3 Back to Main Flow Step 1.
Alternate Flow – No Search Results	2.1 The entered contact number does not match any patient record. 2.2 System displays “No results found”. 2.3 Back to Main Flow Step 3.
Alternate Flow – Not Logged In	3.1 User tries to access patient list without login. 3.2 System redirects user to login page.
Rules	i. Admin must be logged in. ii. Search must be performed using patient contact number only. iii. Search results must be accurate and filtered correctly.
Note	Search is strictly based on the patient’s contact number.

Table 2.2.16.5a View & Search Patient List Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCOV-16-018	Main flow	TCOV-16-023	Patient list and search results displayed	Contact: 0123456789
TCOV-16-019	Alternate Flow – No records	TCOV-16-024	Empty list displayed	0 records
TCOV-16-020	Alternate Flow – No results	TCOV-16-025	No results message	Contact: 0199999999
TCOV-16-021	Alternate Flow – Not logged in	TCOV-16-026	Redirect to login	Direct access without login

Table 2.2.16.5b View & Search Patient List Use Case Test Conditions & Coverage

2.2.17 F017 View & Search Appointment Details

2.2.17.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCOV-17-001	Admin is logged in and	TCOV-17-001	All appointment	5 appointment records

	appointment records exist		records displayed	
TCON-17-002	Search by valid patient contact number	TCOV-17-002	Matching appointment history displayed	Contact: 0123456789
TCON-17-003	Search by unmatched contact number	TCOV-17-003	No appointment found message	Contact: 0199999999
TCON-17-004	Admin is not logged in	TCOV-17-004	Redirect to login page	Direct URL access
TCON-17-005	Appointment status exists	TCOV-17-005	Status displayed correctly	Active / Cancelled by Patient / Cancelled by Doctor

Table 2.2.17.1 View & Search Appointment Details Equivalence Partitioning Test Conditions & Coverage

2.2.17.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-17-006	Number of appointment records = 0	TCOV-17-006	No details to display	0 records
TCON-17-007	Number of appointment records = 1	TCOV-17-007	Single detail view	1 record
TCON-17-008	Number of appointment records > 1	TCOV-17-008	Multiple selectable records	5 records
TCON-17-009	Appointment ID = first record	TCOV-17-009	Boundary record accessed	First appointment
TCON-17-010	Appointment ID = last record	TCOV-17-010	Boundary record accessed	Last appointment

Table 2.2.17.2 View & Search Appointment Details Boundary Value Analysis Test Conditions & Coverage

2.2.17.3 Decision Table Testing

Test Condition ID	Test Condition
TCON-17-011	Admin is logged in
TCON-17-012	Appointment exists

TCOV-17-013	Appointment does not exist
TCOV-17-014	Valid appointment ID
TCOV-17-015	Invalid appointment ID

Table 2.2.17.3a View & Search Appointment Details Decision Table Test Conditions

Condition	TCOV-17-011	TCOV-17-012	TCOV-17-013	TCOV-17-014
Admin Logged In	Y	Y	Y	N
Appointment Exists	Y	Y	N	-
Valid Appointment ID	Y	N	-	-
Output				
Display Appointment Details	Y	N	N	N
Display Error / No Data	N	Y	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.17.3b View & Search Appointment Details Decision Table

2.2.17.4 State Transition Testing

State	Select Appointment (SA)	View Details (VD)	Valid Record (VR)	Invalid Record (IR)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/View Appointment List	S1/-	S1/-	S1/-	S5/Login Page
Appointment List (S2)	S2/-	S3/Load Details	S2/-	S2/-	S5/Login Page
Details Page (S3)	S3/-	S3/-	S4/Display Details	S4/Error Message	S5/Login Page
Result Displayed (S4)	S2/Back to List	S4/-	S4/-	S4/-	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.17.4a View & Search Appointment Details State Table

Test Coverage ID	Test Coverage
TCOV-17-015	S1 to S2 with input SA
TCOV-17-016	S2 to S3 with input VD
TCOV-17-017	S3 to S4 with input VR

TCOV-17-018	S3 to S4 with input IR
TCOV-17-019	S1 to S5 with input NLI

Table 2.2.17.4b View & Search Appointment Details State Coverage

2.2.17.5 Use Case Testing

Use Case ID	UC017
Use Case	F017 View & Search Appointment Details
Purpose	To provide admin with oversight of all scheduled, active, and cancelled hospital appointments.
Actor	Admin
Trigger	Admin clicks the Appointment Details tab.
Precondition	Admin is logged in.
Main Flow	1. Admin selects the Appointment Details tab. 2. System displays all appointments, including status. 3. Admin searches by patient Contact number. 4. System returns the specific appointment history associated with that contact number.
Alternate Flow – No Search Result	1.1 Contact number yields no appointment records. 1.2 System displays an error alert. 1.3 System redirects back to the dashboard.
Rules	i. Admin must be logged in. ii. Appointment search is based on patient Contact number. iii. Appointment status is determined based on userStatus and doctorStatus database flags.
Note	This function is for viewing and searching appointment details only.

Table 2.2.17.5a View & Search Appointment Details Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCOV-17-016	Main flow	TCOV-17-020	Appointment details displayed	Appointment ID: A001
TCOV-17-017	Alternate Flow – Invalid appointment	TCOV-17-021	Error displayed	Appointment ID: invalid
TCOV-17-018	Alternate Flow – Not logged in	TCOV-17-022	Redirect to login	Direct access without login

Table 2.2.17.5b View & Search Appointment Details Use Case Test Conditions & Coverage

2.2.18 F018 View Prescription List

2.2.18.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-18-001	Admin is logged in and prescription records exist	TCOV-18-001	Prescription list displayed	5 prescription records
TCON-18-002	Admin is logged in but no prescription records exist	TCOV-18-002	Empty prescription list displayed	0 prescription records
TCON-18-003	Admin is not logged in	TCOV-18-003	Access denied / redirect to login	Direct URL access
TCON-18-004	Prescription record is valid	TCOV-18-004	Prescription details displayed correctly	Disease: Flu, Allergy: None, Prescription: Paracetamol
TCON-18-005	Prescription record is invalid or missing	TCOV-18-005	Error or empty result handled	Invalid prescription ID

Table 2.2.18.1 View Prescription List Equivalence Partitioning Test Conditions & Coverage

2.2.18.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-18-006	Number of prescription records = 0	TCOV-18-006	Empty list displayed	0 records
TCON-18-007	Number of prescription records = 1	TCOV-18-007	Single prescription displayed	1 record
TCON-18-008	Number of prescription records > 1	TCOV-18-008	Multiple prescriptions displayed	10 records
TCON-18-009	Prescription ID = first record	TCOV-18-009	First prescription record displayed	First prescription record
TCON-18-010	Prescription ID = last record	TCOV-18-010	Last prescription record displayed	Last prescription record

Table 2.2.18.2 View Prescription List Boundary Value Analysis Test Conditions & Coverage

2.2.18.3 Decision Table Testing

Test Condition ID	Test Condition
TCOV-18-011	Admin is logged in
TCOV-18-012	Prescription records exist
TCOV-18-013	Prescription records do not exist
TCOV-18-014	Prescription record is valid
TCOV-18-015	Prescription record is invalid

Table 2.2.18.3a View Prescription List Decision Table Test Conditions

Condition	TCOV-18-011	TCOV-18-012	TCOV-18-013	TCOV-18-014
Admin Logged In	Y	Y	Y	N
Prescription Records Exist	Y	Y	N	-
Prescription Record Valid	Y	N	-	-
Output				
Display Prescription List	Y	N	N	N
Display Error / Invalid Record Message	N	Y	N	N
Display Empty List / Message	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.18.3b View Prescription List Decision Table

2.2.18.4 State Transition Testing

State	Select Prescription List (SPL)	Records Found (RF)	No Records (NR)	Invalid Record (IR)	Not Logged In (NLI)
Admin Dashboard (S1)	S2/Retrieve Prescriptions	S1/-	S1/-	S1/-	S5/Login Page
Retrieve Prescription Records (S2)	S2/-	S3/Display List	S4/Display Empty List	S4/Error Message	S5/Login Page
Prescription List Displayed (S3)	S3/-	S3/-	S3/-	S3/-	S3/-
Empty/Error Displayed (S4)	S1/Back to Dashboard	S4/-	S4/-	S4/-	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.18.4a View Prescription List State Table

Test Coverage ID	Test Coverage
TCOV-18-015	S1 to S2 with input SPL
TCOV-18-016	S2 to S3 with input RF
TCOV-18-017	S2 to S4 with input NR
TCOV-18-018	S2 to S4 with input IR
TCOV-18-019	S1 to S5 with input NLI

Table 2.2.18.4b View Prescription List State Coverage**2.2.18.5 Use Case Testing**

Use Case ID	UC018
Use Case	F018 View Prescription List
Purpose	To allow admin to view all prescription records in the system.
Actor	Admin
Trigger	Admin clicks the Prescription List tab.
Precondition	Admin is logged into the system.
Main Flow	1. Admin selects the Prescription List tab. 2. System retrieves all prescription records from the database. 3. System displays doctor name, patient details, appointment details, disease, allergies, and prescription text in a table. 4. Admin reviews the prescription records.
Alternate Flow – No Prescription Records	1.1 No prescription records exist in the system. 1.2 System displays an empty list or appropriate message. 1.3 Back to Main Flow Step 1.
Alternate Flow – Invalid Record	2.1 System fails to retrieve a valid prescription record. 2.2 System displays an error message. 2.3 Back to Main Flow Step 1.
Alternate Flow – Not Logged In	3.1 User attempts to access the prescription list without admin login. 3.2 System redirects user to the login page.
Rules	i. Admin must be logged into the system. ii. The displayed prescription data is read-only. iii. Only valid prescription records shall be displayed.
Note	Admin can view prescription records but cannot edit prescription details from this function.

Table 2.2.18.5a View Prescription List Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
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TCON-18-016	Main flow	TCOV-18-020	Prescription list displayed	Admin logged in, prescription records exist
TCON-18-017	Alternate Flow – No prescription records	TCOV-18-021	Empty prescription list displayed	0 prescription records
TCON-18-018	Alternate Flow – Invalid record	TCOV-18-022	Error message displayed	Invalid prescription ID
TCON-18-019	Alternate Flow – Not logged in	TCOV-18-023	Redirect to login page	Direct URL access without login

Table 2.2.18.5b View Prescription List Use Case Test Conditions & Coverage

2.2.19 F019 View & Search User Queries

2.2.19.1 Equivalence Partitioning

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-19-001	Admin is logged in and query records exist	TCOV-19-001	Query list displayed	5 query records
TCON-19-002	Admin is logged in but no query records exist	TCOV-19-002	Empty query list displayed	0 records
TCON-19-003	Search by valid contact number	TCOV-19-003	Matching query displayed	Contact: 0123456789
TCON-19-004	Search by unmatched contact number	TCOV-19-004	No results displayed	Contact: 0199999999
TCON-19-005	Admin is not logged in	TCOV-19-005	Redirect to login page	Direct URL access

Table 2.2.19.1 View & Search User Queries Equivalence Partitioning Test Conditions & Coverage

2.2.19.2 Boundary Value Analysis

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Example of Test Data
TCON-19-006	Number of query records = 0	TCOV-19-006	Empty list displayed	0 records

TCOV-19-007	Number of query records = 1	TCOV-19-007	Single query displayed	1 record
TCOV-19-008	Number of query records > 1	TCOV-19-008	Multiple queries displayed	10 records
TCOV-19-009	Query ID = first record	TCOV-19-009	First query displayed	First query record
TCOV-19-010	Query ID = last record	TCOV-19-010	Last query displayed	Last query record

Table 2.2.19.2 View & Search User Queries Boundary Value Analysis Test Conditions & Coverage

2.2.19.3 Decision Table Testing

Test Condition ID	Test Condition
TCOV-19-011	Admin is logged in
TCOV-19-012	Query records exist
TCOV-19-013	Query records do not exist
TCOV-19-014	Query record is valid
TCOV-19-015	Query record is invalid

Table 2.2.19.3a View & Search User Queries Decision Table Test Conditions

Condition	TCOV-19-011	TCOV-19-012	TCOV-19-013	TCOV-19-014
Admin Logged In	Y	Y	Y	N
Query Records Exist	Y	Y	N	-
Query Record Valid	Y	N	-	-
Output				
Display Query List	Y	N	N	N
Display Error Message	N	Y	N	N
Display Empty List	N	N	Y	N
Redirect to Login Page	N	N	N	Y

Table 2.2.19.3b View & Search User Queries Decision Table

2.2.19.4 State Transition Testing

State	Select Query List (SQL)	Records Found (RF)	No Records (NR)	Invalid Record (IR)	Not Logged In (NLI)
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Admin Dashboard (S1)	S2/Retrieve Queries	S1/-	S1/-	S1/-	S5/Login Page
Retrieve Queries (S2)	S2/-	S3/Display List	S4/Display Empty List	S4/Error Message	S5/Login Page
Query List Displayed (S3)	S3/-	S3/-	S3/-	S3/-	S3/-
Empty/Error Display (S4)	S1/Back to Dashboard	S4/-	S4/-	S4/-	S5/Login Page
Login Page (S5)	S5/-	S5/-	S5/-	S5/-	S5/-

Table 2.2.19.4a View & Search User Queries State Table

Test Coverage ID	Test Coverage
TCOV-19-015	S1 to S2 with input SQL
TCOV-19-016	S2 to S3 with input RF
TCOV-19-017	S2 to S4 with input NR
TCOV-19-018	S2 to S4 with input IR
TCOV-19-019	S1 to S5 with input NLI

Table 2.2.19.4b View & Search User Queries State Coverage

2.2.19.5 Use Case Testing

Use Case ID	UC019
Use Case	F019 View & Search User Queries
Purpose	To allow admin to view and search user-submitted queries.
Actor	Admin
Trigger	Admin selects the User Queries tab or submits a search query.
Precondition	Admin is logged into the system.
Main Flow	- Admin selects User Queries. - System retrieves all query records. - System displays name, email, contact number, and message of users. - Admin enters a contact number into the search field. - System displays matching submitted query records.
Alternate Flow – No Query Records	1.1 No queries exist. 1.2 System displays empty list. 1.3 Back to Main Flow Step 1.

Alternate Flow – No Search Result	2.1 Search contact number does not match any query. 2.2 System displays no result message. 2.3 Back to Main Flow Step 4.
Alternate Flow – Not Logged In	3.1 User attempts access without login. 3.2 System redirects to login page.
Rules	i. Admin must be logged in. ii. Query data is read-only. iii. Search is based on contact number.
Note	Queries are submitted by users through the Contact page.

Table 2.2.19.5a View & Search User Queries Use Case

Test Condition ID	Test Condition	Test Coverage ID	Test Coverage	Test Data
TCON-19-016	Main flow	TCOV-19-020	Query list displayed	Admin logged in, queries exist
TCON-19-017	Alternate Flow – No records	TCOV-19-021	Empty list displayed	0 queries
TCON-19-018	Alternate Flow – Invalid record	TCOV-19-022	Error handled	Invalid query ID
TCON-19-019	Alternate Flow – Not logged in	TCOV-19-023	Redirect to login	Direct access without login

Table 2.2.19.5b View & Search User Queries Use Case Test Conditions & Coverage

2.3 Test Identification

2.3.1 F001 Register Account

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-01-001	Register account with all valid inputs	TCOV-01-001	TCON-01-001
TC-01-002	Register account with invalid email format	TCOV-01-002	TCON-01-002
TC-01-003	Register account with password less than 6 characters	TCOV-01-003	TCON-01-003
TC-01-004	Register account with password and confirm password mismatch	TCOV-01-004	TCON-01-004
TC-01-005	Register account with required field empty	TCOV-01-005	TCON-01-005
TC-01-006	Verify password boundary value of 0 characters	TCOV-01-008	TCON-01-008
TC-01-007	Verify password boundary value of 5 characters	TCOV-01-009	TCON-01-009
TC-01-008	Verify password boundary value of 6 characters	TCOV-01-010	TCON-01-010
TC-01-009	Verify password boundary value of 7 characters	TCOV-01-011	TCON-01-011
TC-01-010	Register account main flow	TCOV-01-024	TCON-01-024
TC-01-011	Register account alternate flow – empty field	TCOV-01-025	TCON-01-025
TC-01-012	Register account alternate flow – password mismatch	TCOV-01-026	TCON-01-026
TC-01-013	Register account alternate flow – password fewer than 6 characters	TCOV-01-027	TCON-01-027
TC-01-014	Register account alternate flow – invalid email format	TCOV-01-028	TCON-01-028

Table 2.3.1 Register Account Test Identification**2.3.2 F002 Login**

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-02-001	Login with valid patient credentials	TCOV-02-001	TCON-02-001
TC-02-002	Login with valid doctor credentials	TCOV-02-002	TCON-02-002
TC-02-003	Login with valid admin credentials	TCOV-02-003	TCON-02-003
TC-02-004	Login with non-existing account	TCOV-02-004	TCON-02-004
TC-02-005	Login with incorrect password	TCOV-02-005	TCON-02-005
TC-02-006	Login with empty input fields	TCOV-02-006	TCON-02-006
TC-02-007	Verify password boundary value of 0 characters	TCOV-02-007	TCON-02-007
TC-02-008	Verify password boundary value of 5 characters	TCOV-02-008	TCON-02-008
TC-02-009	Verify password boundary value of 6 characters	TCOV-02-009	TCON-02-009
TC-02-010	Verify password boundary value of 7 characters	TCOV-02-010	TCON-02-010
TC-02-011	Login main flow – patient	TCOV-02-026	TCON-02-021
TC-02-012	Login main flow – doctor	TCOV-02-027	TCON-02-022
TC-02-013	Login main flow – admin	TCOV-02-028	TCON-02-023
TC-02-014	Login alternate flow – empty field	TCOV-02-029	TCON-02-024
TC-02-015	Login alternate flow – invalid credentials	TCOV-02-030	TCON-02-025
TC-02-016	Login alternate flow – wrong role login	TCOV-02-031	TCON-02-026

Table 2.3.2 Login Test Identification

2.3.3 F003 Logout

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-03-001	Logout with active patient session	TCOV-03-001	TCON-03-001
TC-03-002	Logout with active doctor session	TCOV-03-001	TCON-03-001
TC-03-003	Logout with active admin session	TCOV-03-001	TCON-03-001
TC-03-004	Attempt logout without active session	TCOV-03-002	TCON-03-002
TC-03-005	Click "Back to Login Page" after logout	TCOV-03-003	TCON-03-003
TC-03-006	Logout immediately after login (boundary)	TCOV-03-004	TCON-03-004
TC-03-007	Logout after normal usage (boundary)	TCOV-03-005	TCON-03-005
TC-03-008	Attempt logout after session ended	TCOV-03-006	TCON-03-006
TC-03-009	Logout decision – valid session and logout clicked	TCOV-03-007	TCON-03-007
TC-03-010	Logout decision – logout not clicked	TCOV-03-008	TCON-03-008
TC-03-011	Logout decision – no active session	TCOV-03-009	TCON-03-010
TC-03-012	State transition – Dashboard to Logout Page	TCOV-03-010	TCON-03-007
TC-03-013	State transition – Logout Page to Logged Out Message	TCOV-03-011	TCON-03-009
TC-03-014	State transition – Logged Out to Login Page	TCOV-03-012	TCON-03-003
TC-03-015	State transition – No session redirect to Login Page	TCOV-03-013	TCON-03-010
TC-03-016	Logout main flow	TCOV-03-014	TCON-03-011
TC-03-017	Logout alternate flow – no active session	TCOV-03-017	TCON-03-014

TC-03-018	Logout alternate flow – browser back access blocked	TCOV-03-018	TCON-03-015
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Table 2.3.3 Logout Test Identification

2.3.4 F004 Book Appointment

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-04-001	Book appointment with all valid inputs	TCOV-04-001	TCON-04-001
TC-04-002	Book appointment without selecting specialization	TCOV-04-002	TCON-04-002
TC-04-003	Book appointment without selecting doctor	TCOV-04-003	TCON-04-003
TC-04-004	Book appointment without selecting date	TCOV-04-004	TCON-04-004
TC-04-005	Book appointment without selecting time	TCOV-04-005	TCON-04-005
TC-04-006	Book appointment with past date/time	TCOV-04-006	TCON-04-006
TC-04-007	Book appointment when doctor is unavailable	TCOV-04-007	TCON-04-007
TC-04-008	Verify appointment date boundary – past date	TCOV-04-008	TCON-04-008
TC-04-009	Verify appointment date boundary – current date with past time	TCOV-04-009	TCON-04-009
TC-04-010	Verify appointment date boundary – future date	TCOV-04-010	TCON-04-010
TC-04-011	Verify appointment time boundary – earliest slot	TCOV-04-011	TCON-04-011
TC-04-012	Verify appointment time boundary – latest slot	TCOV-04-012	TCON-04-012
TC-04-013	Verify appointment time outside allowed slot	TCOV-04-013	TCON-04-013

TC-04-014	Decision – valid booking conditions satisfied	TCOV-04-014	TCON-04-014
TC-04-015	Decision – missing required fields	TCOV-04-015	TCON-04-019
TC-04-016	Decision – invalid date/time	TCOV-04-016	TCON-04-016
TC-04-017	Decision – doctor unavailable	TCOV-04-017	TCON-04-018
TC-04-018	State transition – Dashboard to Booking Page	TCOV-04-018	TCON-04-014
TC-04-019	State transition – Select specialization to doctor list	TCOV-04-019	TCON-04-014
TC-04-020	State transition – Select doctor to date/time selection	TCOV-04-020	TCON-04-014
TC-04-021	State transition – Select date/time to validation	TCOV-04-021	TCON-04-014
TC-04-022	State transition – Valid booking to success	TCOV-04-023	TCON-04-014
TC-04-023	State transition – Invalid booking to error	TCOV-04-024	TCON-04-019
TC-04-024	Book appointment main flow	TCOV-04-026	TCON-04-020
TC-04-025	Alternate flow – doctor unavailable	TCOV-04-027	TCON-04-021
TC-04-026	Alternate flow – invalid date/time	TCOV-04-028	TCON-04-022
TC-04-027	Alternate flow – missing required field	TCOV-04-029	TCON-04-023

Table 2.3.4 Book Appointment Test Identification

2.3.5 F005 View Appointment History

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-05-001	View appointment history with existing records	TCOV-05-001	TCON-05-001
TC-05-002	View appointment history with no records	TCOV-05-002	TCON-05-002

TC-05-003	Attempt access without login	TCOV-05-003	TCON-05-003
TC-05-004	View active appointment in history	TCOV-05-004	TCON-05-004
TC-05-005	View cancelled appointment in history	TCOV-05-005	TCON-05-005
TC-05-006	Boundary – 0 appointment records	TCOV-05-006	TCON-05-006
TC-05-007	Boundary – 1 appointment record	TCOV-05-007	TCON-05-007
TC-05-008	Boundary – multiple appointment records	TCOV-05-008	TCON-05-008
TC-05-009	Boundary – past appointment displayed	TCOV-05-009	TCON-05-009
TC-05-010	Boundary – future appointment displayed	TCOV-05-010	TCON-05-010
TC-05-011	Decision – logged in with records	TCOV-05-011	TCON-05-011
TC-05-012	Decision – logged in without records	TCOV-05-013	TCON-05-013
TC-05-013	Decision – not logged in	TCOV-05-014	TCON-05-014
TC-05-014	State transition – Dashboard to history page	TCOV-05-015	TCON-05-011
TC-05-015	State transition – records found	TCOV-05-016	TCON-05-012
TC-05-016	State transition – no records	TCOV-05-017	TCON-05-013
TC-05-017	State transition – not logged in redirect	TCOV-05-020	TCON-05-014
TC-05-018	Main flow – view appointment history	TCOV-05-021	TCON-05-016
TC-05-019	Alternate flow – no appointment records	TCOV-05-022	TCON-05-017
TC-05-020	Alternate flow – active appointment display	TCOV-05-023	TCON-05-018

TC-05-021	Alternate flow – cancelled appointment display	TCOV-05-024	TCON-05-019
TC-05-022	Alternate flow – not logged in	TCOV-05-025	TCON-05-020

Table 2.3.5 View Appointment History Test Identification

2.3.6 F006 Cancel Appointment

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-06-001	Cancel appointment with active status	TCOV-06-001	TCON-06-001
TC-06-002	Attempt to cancel already cancelled appointment	TCOV-06-002	TCON-06-002
TC-06-003	Confirm cancellation (OK clicked)	TCOV-06-003	TCON-06-003
TC-06-004	Reject cancellation (Cancel clicked)	TCOV-06-004	TCON-06-004
TC-06-005	Attempt cancellation without login	TCOV-06-005	TCON-06-005
TC-06-006	Boundary – 0 active appointments	TCOV-06-006	TCON-06-006
TC-06-007	Boundary – 1 active appointment	TCOV-06-007	TCON-06-007
TC-06-008	Boundary – multiple active appointments	TCOV-06-008	TCON-06-008
TC-06-009	Boundary – status transition Active → Cancelled	TCOV-06-009	TCON-06-009
TC-06-010	Decision – valid cancellation conditions met	TCOV-06-010	TCON-06-010
TC-06-011	Decision – cancellation rejected by user	TCOV-06-011	TCON-06-015
TC-06-012	Decision – appointment not active	TCOV-06-012	TCON-06-013
TC-06-013	Decision – not logged in	TCOV-06-013	TCON-06-010

TC-06-014	State transition – Dashboard to appointment history	TCOV-06-014	TCON-06-010
TC-06-015	State transition – click cancel to confirmation dialog	TCOV-06-015	TCON-06-010
TC-06-016	State transition – confirm cancellation to status update	TCOV-06-016	TCON-06-014
TC-06-017	State transition – reject cancellation back to history	TCOV-06-017	TCON-06-015
TC-06-018	State transition – invalid cancellation to error state	TCOV-06-018	TCON-06-013
TC-06-019	State transition – not logged in redirect	TCOV-06-019	TCON-06-019
TC-06-020	Main flow – cancel appointment successfully	TCOV-06-020	TCON-06-016
TC-06-021	Alternate flow – cancellation rejected	TCOV-06-021	TCON-06-017
TC-06-022	Alternate flow – already cancelled appointment	TCOV-06-022	TCON-06-018
TC-06-023	Alternate flow – not logged in	TCOV-06-023	TCON-06-019

Table 2.3.6 Cancel Appointment Test Identification

2.3.7 F007 View Prescription

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-07-001	View prescription with existing records	TCOV-07-001	TCON-07-001
TC-07-002	View prescription with no records	TCOV-07-002	TCON-07-002
TC-07-003	Attempt access without login	TCOV-07-003	TCON-07-003
TC-07-004	Click Pay Bill button	TCOV-07-004	TCON-07-004
TC-07-005	Bill payment successful	TCOV-07-005	TCON-07-005
TC-07-006	Bill payment failed	TCOV-07-006	TCON-07-006

TC-07-007	Generate PDF after successful payment	TCOV-07-007	TCON-07-007
TC-07-008	Boundary – 0 prescription records	TCOV-07-008	TCON-07-008
TC-07-009	Boundary – 1 prescription record	TCOV-07-009	TCON-07-009
TC-07-010	Boundary – multiple prescription records	TCOV-07-010	TCON-07-010
TC-07-011	Attempt PDF generation before payment	TCOV-07-011	TCON-07-011
TC-07-012	Attempt PDF generation after payment	TCOV-07-012	TCON-07-012
TC-07-013	Main flow – view, pay, generate PDF	TCOV-07-024	TCON-07-018
TC-07-014	Alternate flow – no prescription	TCOV-07-025	TCON-07-019
TC-07-015	Alternate flow – payment failed	TCOV-07-026	TCON-07-020

Table 2.3.7 View Prescription Test Identification

2.3.8 F008 Submit Contact Query

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-08-001	Submit query with all valid inputs	TCOV-08-001	TCON-08-001
TC-08-002	Submit query with empty name	TCOV-08-002	TCON-08-002
TC-08-003	Submit query with invalid email	TCOV-08-003	TCON-08-003
TC-08-004	Submit query with empty phone number	TCOV-08-004	TCON-08-004
TC-08-005	Submit query with invalid phone number	TCOV-08-005	TCON-08-005
TC-08-006	Submit query with empty message	TCOV-08-006	TCON-08-006
TC-08-007	Submit query with all fields empty	TCOV-08-007	TCON-08-007
TC-08-008	Boundary – phone length = 0	TCOV-08-008	TCON-08-008

TC-08-009	Boundary – phone contains letters	TCOV-08-009	TCON-08-009
TC-08-010	Boundary – valid phone number	TCOV-08-010	TCON-08-010
TC-08-011	Boundary – message length = 0	TCOV-08-011	TCON-08-011
TC-08-012	Boundary – message length = 1	TCOV-08-012	TCON-08-012
TC-08-013	Main flow – submit contact query	TCOV-08-019	TCON-08-019
TC-08-014	Alternate flow – missing field	TCOV-08-020	TCON-08-020
TC-08-015	Alternate flow – invalid email	TCOV-08-021	TCON-08-021
TC-08-016	Alternate flow – invalid phone	TCOV-08-022	TCON-08-022

Table 2.3.8 Submit Contact Query Test Identification

2.3.9 F009 View Appointment

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-09-001	View appointment list with records	TCOV-09-001	TCON-09-001
TC-09-002	View appointment list with no records	TCOV-09-002	TCON-09-002
TC-09-003	Search appointment by valid contact number	TCOV-09-003	TCON-09-003
TC-09-004	Search appointment with no matching contact number	TCOV-09-004	TCON-09-004
TC-09-005	Attempt access without login	TCOV-09-005	TCON-09-005
TC-09-006	Boundary – 0 appointments	TCOV-09-006	TCON-09-006
TC-09-007	Boundary – 1 appointment	TCOV-09-007	TCON-09-007
TC-09-008	Boundary – multiple appointments	TCOV-09-008	TCON-09-008
TC-09-009	Main flow – view and search appointment	TCOV-09-020	TCON-09-016

TC-09-010	Alternate flow – no appointment	TCOV-09-021	TCON-09-017
TC-09-011	Alternate flow – no search result	TCOV-09-022	TCON-09-018
TC-09-012	Alternate flow – not logged in	TCOV-09-019	TCON-09-005

Table 2.3.9 View Appointment Test Identification

2.3.10 F010 Prescribe Medication

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-10-001	Prescribe medication with valid inputs	TCOV-10-001	TCON-10-001
TC-10-002	Submit with empty disease field	TCOV-10-002	TCON-10-002
TC-10-003	Submit with empty prescription field	TCOV-10-003	TCON-10-003
TC-10-004	Submit with invalid appointment ID	TCOV-10-004	TCON-10-004
TC-10-005	Submit when appointment is not active	TCOV-10-005	TCON-10-005
TC-10-006	Attempt access without login	TCOV-10-006	TCON-10-006
TC-10-007	Boundary – disease length = 0	TCOV-10-007	TCON-10-007
TC-10-008	Boundary – disease length = 1	TCOV-10-008	TCON-10-008
TC-10-009	Boundary – prescription length = 0	TCOV-10-009	TCON-10-009
TC-10-010	Boundary – prescription length = 1	TCOV-10-010	TCON-10-010
TC-10-011	Boundary – allergy field empty (optional)	TCOV-10-011	TCON-10-011
TC-10-012	Boundary – allergy field provided	TCOV-10-012	TCON-10-012
TC-10-013	Decision – all conditions valid	TCOV-10-013	TCON-10-013
TC-10-014	Decision – missing required fields	TCOV-10-014	TCON-10-018
TC-10-015	Decision – invalid/inactive appointment	TCOV-10-015	TCON-10-019

TC-10-016	Decision – not logged in	TCOV-10-016	TCON-10-013
TC-10-017	State transition – appointment list to prescription form	TCOV-10-017	TCON-10-013
TC-10-018	State transition – input entered to validation	TCOV-10-019	TCON-10-013
TC-10-019	State transition – valid input to save prescription	TCOV-10-020	TCON-10-013
TC-10-020	State transition – invalid input to error message	TCOV-10-021	TCON-10-018
TC-10-021	State transition – invalid appointment to error state	TCOV-10-022	TCON-10-019
TC-10-022	State transition – save success to confirmation	TCOV-10-023	TCON-10-013
TC-10-023	Main flow – prescribe medication successfully	TCOV-10-025	TCON-10-020
TC-10-024	Alternate flow – missing required fields	TCOV-10-026	TCON-10-021
TC-10-025	Alternate flow – invalid or inactive appointment	TCOV-10-027	TCON-10-022
TC-10-026	Alternate flow – save failed	TCOV-10-028	TCON-10-023

Table 2.3.10 Prescribe Medication Test Identification

2.3.11 F011 Cancel Appointment

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-11-001	Cancel appointment with active status (assigned doctor)	TCOV-11-001	TCON-11-001
TC-11-002	Attempt to cancel already cancelled appointment	TCOV-11-002	TCON-11-002
TC-11-003	Confirm cancellation (OK clicked)	TCOV-11-003	TCON-11-003

TC-11-004	Reject cancellation (Cancel clicked)	TCOV-11-004	TCON-11-004
TC-11-005	Attempt to cancel appointment not assigned to doctor	TCOV-11-005	TCON-11-005
TC-11-006	Attempt cancellation without login	TCOV-11-006	TCON-11-006
TC-11-007	Boundary – 0 active appointments	TCOV-11-007	TCON-11-007
TC-11-008	Boundary – 1 active appointment	TCOV-11-008	TCON-11-008
TC-11-009	Boundary – multiple active appointments	TCOV-11-009	TCON-11-009
TC-11-010	Boundary – status transition Active → Cancelled	TCOV-11-010	TCON-11-010
TC-11-011	Decision – valid cancellation conditions met	TCOV-11-011	TCON-11-011
TC-11-012	Decision – cancellation rejected by user	TCOV-11-012	TCON-11-016
TC-11-013	Decision – appointment not active	TCOV-11-013	TCON-11-013
TC-11-014	Decision – not assigned to doctor	TCOV-11-014	TCON-11-014
TC-11-015	Decision – not logged in	TCOV-11-015	TCON-11-011
TC-11-016	State transition – dashboard to appointment list	TCOV-11-016	TCON-11-011
TC-11-017	State transition – click cancel to confirmation dialog	TCOV-11-017	TCON-11-011
TC-11-018	State transition – confirm cancellation to update status	TCOV-11-018	TCON-11-015
TC-11-019	State transition – reject cancellation back to list	TCOV-11-019	TCON-11-016
TC-11-020	State transition – not assigned to error state	TCOV-11-020	TCON-11-014

TC-11-021	State transition – not logged in redirect	TCOV-11-021	TCON-11-011
TC-11-022	Main flow – cancel appointment successfully	TCOV-11-022	TCON-11-017
TC-11-023	Alternate flow – reject confirmation	TCOV-11-023	TCON-11-018
TC-11-024	Alternate flow – not assigned appointment	TCOV-11-024	TCON-11-019
TC-11-025	Alternate flow – not logged in	TCOV-11-025	TCON-11-020

Table 2.3.11 Cancel Appointment Test Identification

2.3.12 F012 View Prescription List

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-12-001	View prescription list with existing records	TCOV-12-001	TCON-12-001
TC-12-002	View prescription list with no records	TCOV-12-002	TCON-12-002
TC-12-003	Attempt access without login	TCOV-12-003	TCON-12-003
TC-12-004	View valid prescription details	TCOV-12-004	TCON-12-004
TC-12-005	Handle invalid/missing prescription record	TCOV-12-005	TCON-12-005
TC-12-006	Boundary – 0 prescription records	TCOV-12-006	TCON-12-006
TC-12-007	Boundary – 1 prescription record	TCOV-12-007	TCON-12-007
TC-12-008	Boundary – multiple prescription records	TCOV-12-008	TCON-12-008
TC-12-009	Boundary – prescription belongs to logged-in doctor	TCOV-12-009	TCON-12-009
TC-12-010	Boundary – prescription not belonging to logged-in doctor	TCOV-12-010	TCON-12-010

TC-12-011	Decision – logged in with valid records	TCOV-12-011	TCON-12-011
TC-12-012	Decision – records exist but unauthorized	TCOV-12-012	TCON-12-014
TC-12-013	Decision – no records exist	TCOV-12-013	TCON-12-013
TC-12-014	Decision – not logged in	TCOV-12-014	TCON-12-011
TC-12-015	State transition – dashboard to retrieve prescriptions	TCOV-12-015	TCON-12-011
TC-12-016	State transition – records found to display list	TCOV-12-016	TCON-12-012
TC-12-017	State transition – no records to empty display	TCOV-12-017	TCON-12-013
TC-12-018	State transition – unauthorized record hidden	TCOV-12-018	TCON-12-015
TC-12-019	State transition – not logged in redirect	TCOV-12-019	TCON-12-011
TC-12-020	Main flow – view prescription list	TCOV-12-020	TCON-12-016
TC-12-021	Alternate flow – no prescription records	TCOV-12-021	TCON-12-017
TC-12-022	Alternate flow – unauthorized record	TCOV-12-022	TCON-12-018
TC-12-023	Alternate flow – not logged in	TCOV-12-023	TCON-12-019

Table 2.3.12 View Prescription List Test Identification

2.3.13 F013 Add Doctor

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-13-001	Add doctor with all valid inputs	TCOV-13-001	TCON-13-001
TC-13-002	Add doctor with missing required field	TCOV-13-002	TCON-13-002

TC-13-003	Add doctor with invalid email format	TCOV-13-003	TCON-13-003
TC-13-004	Add doctor with password mismatch	TCOV-13-004	TCON-13-004
TC-13-005	Add doctor with invalid consultation fee	TCOV-13-005	TCON-13-005
TC-13-006	Attempt to add doctor without login	TCOV-13-006	TCON-13-006
TC-13-007	Boundary – password length = 0	TCOV-13-007	TCON-13-007
TC-13-008	Boundary – password length = 5 (below minimum)	TCOV-13-008	TCON-13-008
TC-13-009	Boundary – password length = 6 (minimum valid)	TCOV-13-009	TCON-13-009
TC-13-010	Boundary – password length > 6	TCOV-13-010	TCON-13-010
TC-13-011	Boundary – consultation fee = 0	TCOV-13-011	TCON-13-011
TC-13-012	Boundary – consultation fee > 0	TCOV-13-012	TCON-13-012
TC-13-013	Decision – all inputs valid	TCOV-13-013	TCON-13-013
TC-13-014	Decision – missing required fields	TCOV-13-014	TCON-13-018
TC-13-015	Decision – invalid email or fee	TCOV-13-015	TCON-13-003
TC-13-016	Decision – password mismatch	TCOV-13-016	TCON-13-004
TC-13-017	Decision – not logged in	TCOV-13-017	TCON-13-019
TC-13-018	State transition – dashboard to add doctor form	TCOV-13-018	TCON-13-013
TC-13-019	State transition – input entered to validation	TCOV-13-020	TCON-13-013

TC-13-020	State transition – valid input to create record	TCOV-13-021	TCON-13-013
TC-13-021	State transition – invalid input to error message	TCOV-13-022	TCON-13-018
TC-13-022	State transition – not logged in redirect	TCOV-13-023	TCON-13-019
TC-13-023	Main flow – add doctor successfully	TCOV-13-025	TCON-13-020
TC-13-024	Alternate flow – missing required field	TCOV-13-026	TCON-13-021
TC-13-025	Alternate flow – password mismatch	TCOV-13-027	TCON-13-022
TC-13-026	Alternate flow – invalid input	TCOV-13-028	TCON-13-023
TC-13-027	Alternate flow – not logged in	TCOV-13-029	TCON-13-024

Table 2.3.13 Add Doctor Test Identification

2.3.14 F014 Delete Doctor

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-14-001	Delete doctor using existing registered Email ID	TCOV-14-001	TCON-14-001
TC-14-002	Attempt to delete doctor using non-existing Email ID	TCOV-14-002	TCON-14-002
TC-14-003	Confirm deletion using OK button	TCOV-14-003	TCON-14-003
TC-14-004	Reject deletion using Cancel button	TCOV-14-004	TCON-14-004
TC-14-005	Handle deletion query failure	TCOV-14-005	TCON-14-005
TC-14-006	Attempt deletion without admin login	TCOV-14-006	TCON-14-006
TC-14-007	Boundary – 0 doctor records	TCOV-14-007	TCON-14-007
TC-14-008	Boundary – 1 doctor record	TCOV-14-008	TCON-14-008

TC-14-009	Boundary – multiple doctor records	TCOV-14-009	TCON-14-009
TC-14-010	Boundary – existing doctor email entered	TCOV-14-010	TCON-14-010
TC-14-011	Boundary – non-existing doctor email entered	TCOV-14-011	TCON-14-011
TC-14-012	Decision – valid deletion conditions met	TCOV-14-012	TCON-14-012
TC-14-013	Decision – confirmation denied	TCOV-14-013	TCON-14-015
TC-14-014	Decision – doctor email does not exist	TCOV-14-014	TCON-14-013
TC-14-015	Decision – not logged in	TCOV-14-015	TCON-14-012
TC-14-016	State transition – dashboard to delete doctor form	TCOV-14-018	TCON-14-012
TC-14-017	State transition – enter Email ID	TCOV-14-019	TCON-14-013
TC-14-018	State transition – submit delete request to confirmation dialog	TCOV-14-020	TCON-14-014
TC-14-019	State transition – confirm deletion to doctor deleted	TCOV-14-021	TCON-14-016
TC-14-020	State transition – reject deletion back to form	TCOV-14-022	TCON-14-015
TC-14-021	State transition – delete failed to error message	TCOV-14-023	TCON-14-017
TC-14-022	State transition – not logged in redirect	TCOV-14-024	TCON-14-012
TC-14-023	Main flow – delete doctor successfully	TCOV-14-025	TCON-14-018
TC-14-024	Alternate flow – confirmation denied	TCOV-14-026	TCON-14-019
TC-14-025	Alternate flow – delete failed	TCOV-14-027	TCON-14-020

TC-14-026	Alternate flow – not logged in	TCOV-14-028	TCON-14-021
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Table 2.3.14 Delete Doctor Test Identification

2.3.15 F015 View & Search Doctor List

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-15-001	View doctor list with records	TCOV-15-001	TCON-15-001
TC-15-002	View doctor list with no records	TCOV-15-002	TCON-15-002
TC-15-003	Attempt access without login	TCOV-15-003	TCON-15-003
TC-15-004	Search doctor by valid Email ID	TCOV-15-004	TCON-15-004
TC-15-005	Search doctor with no matching Email ID	TCOV-15-005	TCON-15-005
TC-15-006	Empty search input shows full list	TCOV-15-006	TCON-15-006
TC-15-007	Boundary – 0 records	TCOV-15-007	TCON-15-007
TC-15-008	Boundary – 1 record	TCOV-15-008	TCON-15-008
TC-15-009	Boundary – multiple records	TCOV-15-009	TCON-15-009
TC-15-010	Boundary – valid Email ID format	TCOV-15-011	TCON-15-011
TC-15-011	Boundary – invalid Email ID format	TCOV-15-012	TCON-15-012
TC-15-012	Main flow – view and search doctor	TCOV-15-023	TCON-15-018
TC-15-013	Alternate flow – no records	TCOV-15-024	TCON-15-019
TC-15-014	Alternate flow – no results	TCOV-15-025	TCON-15-020
TC-15-015	Alternate flow – not logged in	TCOV-15-026	TCON-15-021

Table 2.3.15 View & Search Doctor List Test Identification

2.3.16 F016 View & Search Patient List

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-16-001	View patient list with records	TCOV-16-001	TCON-16-001
TC-16-002	View patient list with no records	TCOV-16-002	TCON-16-002

TC-16-003	Attempt access without login	TCOV-16-003	TCON-16-003
TC-16-004	Search patient by valid contact number	TCOV-16-004	TCON-16-004
TC-16-005	Search patient with no matching contact number	TCOV-16-005	TCON-16-005
TC-16-006	Empty search input shows full list	TCOV-16-006	TCON-16-006
TC-16-007	Boundary – 0 records	TCOV-16-007	TCON-16-007
TC-16-008	Boundary – 1 record	TCOV-16-008	TCON-16-008
TC-16-009	Boundary – multiple records	TCOV-16-009	TCON-16-009
TC-16-010	Boundary – valid contact number	TCOV-16-011	TCON-16-011
TC-16-011	Boundary – invalid contact number	TCOV-16-012	TCON-16-012
TC-16-012	Main flow – view and search patient	TCOV-16-023	TCON-16-018
TC-16-013	Alternate flow – no records	TCOV-16-024	TCON-16-019
TC-16-014	Alternate flow – no results	TCOV-16-025	TCON-16-020
TC-16-015	Alternate flow – not logged in	TCOV-16-026	TCON-16-021

Table 2.3.16 View & Search Patient List Test Identification

2.3.17 F017 View & Search Appointment Details

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-17-001	View all appointment records	TCOV-17-001	TCON-17-001
TC-17-002	Search appointment by valid contact number	TCOV-17-002	TCON-17-002
TC-17-003	Search appointment with no result	TCOV-17-003	TCON-17-003
TC-17-004	Attempt access without login	TCOV-17-004	TCON-17-004
TC-17-005	Boundary – 0 records	TCOV-17-006	TCON-17-006

TC-17-006	Boundary – 1 record	TCOV-17-007	TCON-17-007
TC-17-007	Boundary – multiple records	TCOV-17-008	TCON-17-008
TC-17-008	Main flow – view and search appointment	TCOV-17-020	TCON-17-016
TC-17-009	Alternate flow – no result	TCOV-17-021	TCON-17-017
TC-17-010	Alternate flow – not logged in	TCOV-17-022	TCON-17-018

Table 2.3.17 View & Search Appointment Details Test Identification

2.3.18 F018 View Prescription List

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-18-001	View prescription list with existing records	TCOV-18-001	TCON-18-001
TC-18-002	View prescription list with no records	TCOV-18-002	TCON-18-002
TC-18-003	Attempt access without login	TCOV-18-003	TCON-18-003
TC-18-004	Display valid prescription details correctly	TCOV-18-004	TCON-18-004
TC-18-005	Handle invalid/missing prescription record	TCOV-18-005	TCON-18-005
TC-18-006	Boundary – 0 prescription records	TCOV-18-006	TCON-18-006
TC-18-007	Boundary – 1 prescription record	TCOV-18-007	TCON-18-007
TC-18-008	Boundary – multiple prescription records	TCOV-18-008	TCON-18-008
TC-18-009	Boundary – first prescription record accessed	TCOV-18-009	TCON-18-009
TC-18-010	Boundary – last prescription record accessed	TCOV-18-010	TCON-18-010
TC-18-011	Decision – logged in with records	TCOV-18-011	TCON-18-011

TC-18-012	Decision – no prescription records	TCOV-18-013	TCON-18-013
TC-18-013	Decision – invalid record	TCOV-18-014	TCON-18-015
TC-18-014	Decision – not logged in	TCOV-18-014	TCON-18-011
TC-18-015	State transition – dashboard to retrieve prescriptions	TCOV-18-015	TCON-18-011
TC-18-016	State transition – records found to display list	TCOV-18-016	TCON-18-012
TC-18-017	State transition – no records to empty display	TCOV-18-017	TCON-18-013
TC-18-018	State transition – invalid record to error display	TCOV-18-018	TCON-18-015
TC-18-019	State transition – not logged in redirect	TCOV-18-019	TCON-18-011
TC-18-020	Main flow – view prescription list	TCOV-18-020	TCON-18-016
TC-18-021	Alternate flow – no prescription records	TCOV-18-021	TCON-18-017
TC-18-022	Alternate flow – invalid record	TCOV-18-022	TCON-18-018
TC-18-023	Alternate flow – not logged in	TCOV-18-023	TCON-18-019

Table 2.3.18 View Prescription List Test Identification

2.3.19 F019 View & Search User Queries

Test Case ID	Description	Test Coverage ID	Test Condition ID
TC-19-001	View user queries with records	TCOV-19-001	TCON-19-001
TC-19-002	View user queries with no records	TCOV-19-002	TCON-19-002
TC-19-003	Search query by valid contact number	TCOV-19-003	TCON-19-003
TC-19-004	Search query with no result	TCOV-19-004	TCON-19-004
TC-19-005	Attempt access without login	TCOV-19-005	TCON-19-005

TC-19-006	Boundary – 0 records	TCOV-19-006	TCON-19-006
TC-19-007	Boundary – 1 record	TCOV-19-007	TCON-19-007
TC-19-008	Boundary – multiple records	TCOV-19-008	TCON-19-008
TC-19-009	Main flow – view and search queries	TCOV-19-020	TCON-19-016
TC-19-010	Alternate flow – no records	TCOV-19-021	TCON-19-017
TC-19-011	Alternate flow – no search result	TCOV-19-022	TCON-19-018
TC-19-012	Alternate flow – not logged in	TCOV-19-023	TCON-19-019

Table 2.3.19 View & Search User Queries Test Identification

2.4 Feature Pass/Fail Criteria

Each feature must satisfy the following criteria in order to pass:

- i. Pass all its test cases
- ii. 0 validation or processing error
- iii. No system failure
- iv. Fulfill its requirement

2.5 Test Deliverables

The following documents will be generated for the testing activities of the system and will be delivered.

Test Documentation:

- i. Software Requirements Specification (SRS)
- ii. Test Plan
- iii. Test Design Specification
- iv. Test Case Specifications
- v. Test Procedure Specifications
- vi. Test Automation Tool / Test Management Tool Report
- vii. Test Logs
- viii. Test Incident Reports
- ix. Test Summary Report
- x. Test Completion Report